Cleaning and Merging Datasets

Supplementary

Data cleaning

Only autosomal SNPs were analysed. SNPs with a missingness rate above 0.05 were removed. Individuals with a genotype call rate less than 0.9 (missingness exceeding 0.1) were removed. SNPs with minor allele frequency below 0.025 were removed. Hardy-Weinberg equilibrium p-values were calculated for each SNP within each population using the tidypopgen function qc_report_loci, which provides a Bonferroni correction given the number of populations analysed. SNPs with a p-value below 0.01 in at least one population after correction were removed. Close relatives were removed using the KING robust kinship coefficient Manichaikul et al. (2010), implemented in tidypopgen. One individual was removed from each pair with a kinship coefficient greater than 0.0884, equivalent to removing relatives up to the second degree. After this filtering, the Malagasy dataset consisted of 233 individuals and 596146 SNPs, and the global dataset consisted of 2744 individuals and 425931 SNPs.

SNPs in linkage were then removed by LD clumping Privé et al. (2018), using tidypopgen. Those with an r2 value greater than 0.2 in a window of 100 kilobases were removed. After LD clumping, the Malagasy dataset consisted of 227410 SNPs and the global dataset consisted of 198196 SNPs.

PCA

Principal components analysis (PCA) was used to visualise the Malagasy dataset, and the Malagasy individuals compared to global comparative data. Non-malagasy populations were randomly subsampled to a maximum of 30 individuals to avoid uneven sampling bias. This left a total of 1955 individuals.

Clustering

Unsupervised clustering using ADMIXTURE Alexander, Novembre, and Lange (2009) was run on the full, cleaned dataset of 2744 individuals and 425931 SNPs using the package tidypopgen E. J. Carter et al. (2025). The algorithm was run for 9 values of K (2 to 10) with 5-fold cross validation.

f_2 and f_3 statistics

f_2 statistics were calculated in tidypopgen E. Carter and Manica (2025), f_3 and f_4 ratio statistics were then calculated in Admixtools2 from these f_2 blocks Maier and Patterson (2024). Admixture f_3 statistics were used in the configuration f_3(Malagasy; Asian, African) to test whether the Malagasy population could be modelled as an admixture between the given African and Asian references.

Merina substructure

As the Merina exhibited high inter-individual variation in ADMIXTURE ancestry profiles, indicative of some form of population structure, we ran supervised ADMIXTURE Alexander, Novembre, and Lange (2009) at

K = 2 for all Malagasy populations. Borneo and Mozambique were used as reference populations in this supervised mode, as these reference populations gave the minimum f3 values. Each population showed inter-individual variation in ancestry proportions. However, the Merina exhibited a bimodal distribution Alexander, Novembre, and Lange (2009), as reported previously Hodgson (2016). Merina with a high proportion of African ancestry and Merina with a high proportion of Asian ancestry were separated before running MALDER and pairwise FST analyses. Individuals were assigned to Merina A or Merina B based on their ancestry profile at k = 2 inferred using supervised ADMIXTURE Alexander, Novembre, and Lange (2009), with Merina A estimated to have >50% of ancestry derived from a population similar to Borneo individuals, and Merina B estimated to have >50% of ancestry derived from a population similar to Mozambique individuals. See supplementary Figure 2 for results of supervised ADMIXTURE analysis. Splitting this sample reduced power, as only 12 individuals remained in the higher African ancestry group. However, this avoided violating model assumptions (e.g the model of linkage employed by ALDER assumes no population substructure and panmixia) and enabled us to consider whether there may be differing demographic histories among the Merina.

f_4 ratio

f_4 ratio statistics were used to examine the proportion of ancestry derived from Asian versus African sources for each population. To obtain these ratios, the global dataset was merged with 1000 Genomes GBR and a chimpanzee reference (PanTro3.0, GenBank accession GCA_000001515.5) to act as outgroups. f_4 ratios were calculated in the configurations f_4(GBR,Chimpanzee;Malagasy,Borneo) / f_4(GBR,Chimpanzee;Esan,Borneo) and f_4(GBR,Chimpanzee;Malagasy,Esan) / f_4(GBR,Chimpanzee;Borneo,Esan).

To obtain a chimpanzee reference, SNPs in our dataset were lifted to PanTro3.0 (GenBank accession GCA_000001515.5) from hg19 using liftOver overchain file https://hgdownload.soe.ucsc.edu/goldenPath/hg19/liftOver/hg19ToPanTro5.over.chain.gz from UCSC Genome Browser Kent et al. (2002), using bigsnpr function snp_modifyBuild() Privé et al. (2018). The chimpanzee reference for each SNP was obtained using a custom R function to query the Ensembl API (GenBank accession GCA_000001515.5, Ensembl release 115, accessed via Ensembl API https://rest.ensembl.org/documentation/info/sequence_region on 17-12-2025). This procedure retained 77993 SNPs for f_4 ratio calculation. Standard errors were calculated using block-jackknife resampling (admixtools f_4 ratio default). Supplementary Table 5 contains full f_4 ratio output.

MALDER

MALDER Loh et al. (2013); Pickrell et al. (2014) was used to date the timing of admixture. Genetic distances were set using HapMap Phase II genetic map Frazer et al. (2007), mapped to build GRCh37. tidypopgen was used to create a subset of the full dataset containing only SNPs with known genetic map positions in the correct order. This dataset was not pruned for linkage disequilibrium, and consisted of 425904 SNPs. MALDER was run using the default -binsize of 5 * 10-4 Morgans (describing the size of the bins that SNPs are split up into. Instead of directly calculating distance between each SNP pair, MALDER uses the distance between the bins of each SNP pair.), and the –mindis parameter (minimum genetic distance to begin LD decay curve fitting) set to 0.005 cM. Standard error of estimated dates were calculated through leave-one-chromosome-out jackknifing, with the flag -jackknife.

Pairwise F_{st}

Pairwise F_{st} distances between Malagasy populations were calculated in tidypopgen, using the method of Weir and Cockerham (1984).

Supplementary Figures

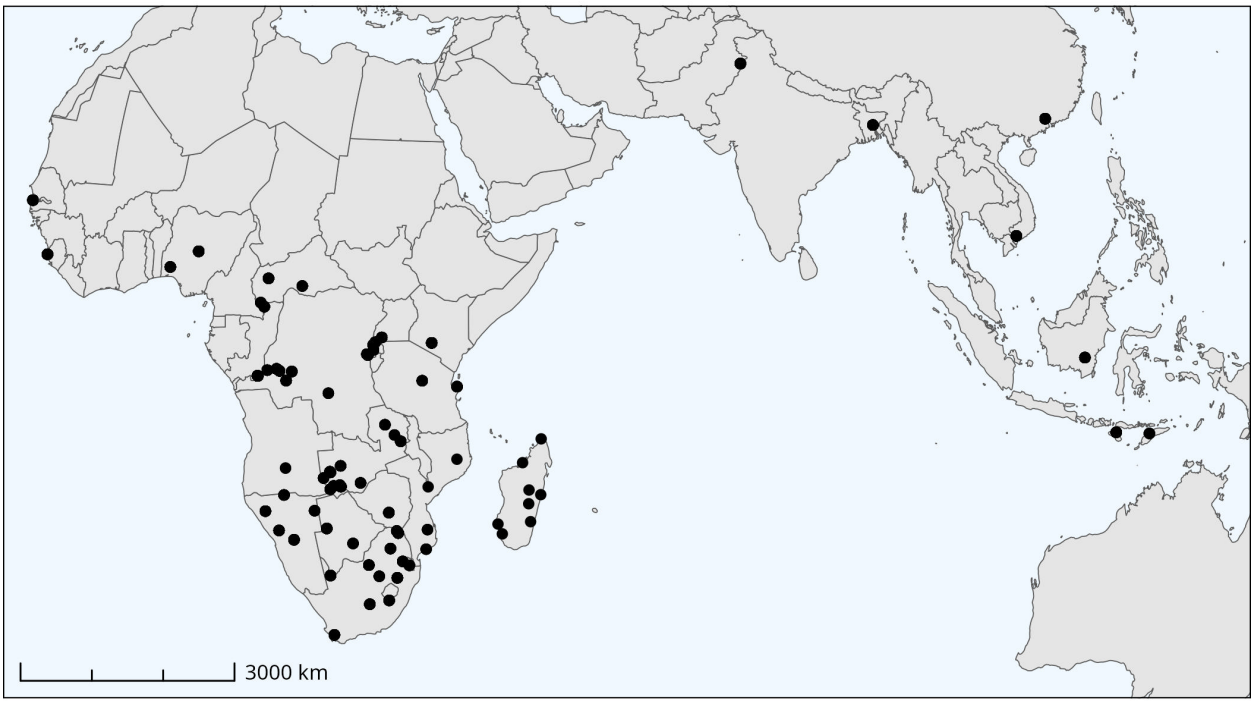


Figure 1: Supplementary Figure 1: Map of reference population locations.

Supplementary Tables

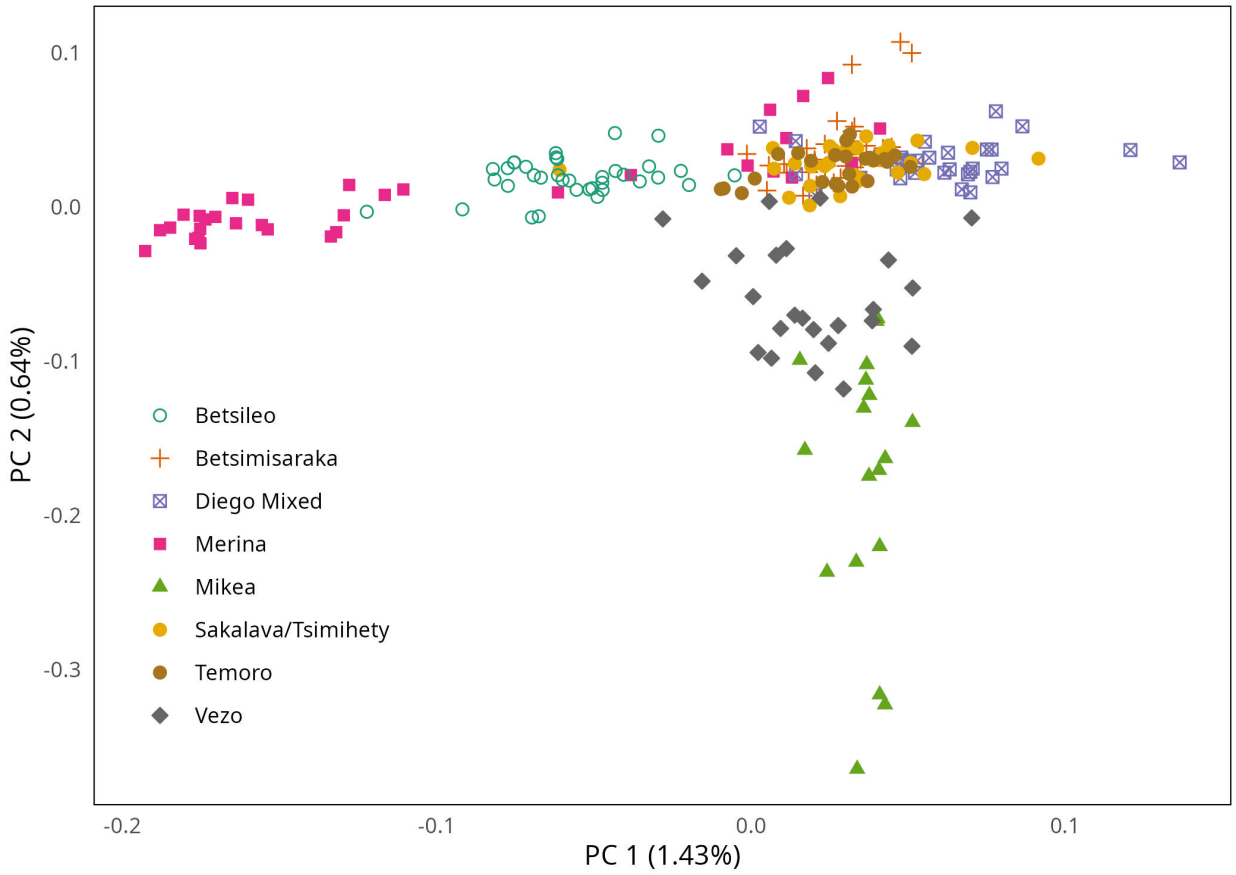


Figure 2: Supplementary Figure 2: Principal components analysis of Malagasy data (n = 167), combined with data from Pierron et al. (2014) (Mikea, Vezo, and Temoro data points, n = 69). Colour and shape indicates self-identified ethnicity. 'Diego Mixed' individuals identified as Anjoaty, Antakarana, Sakalava Tavaratra, and Sakalava ethnicities.

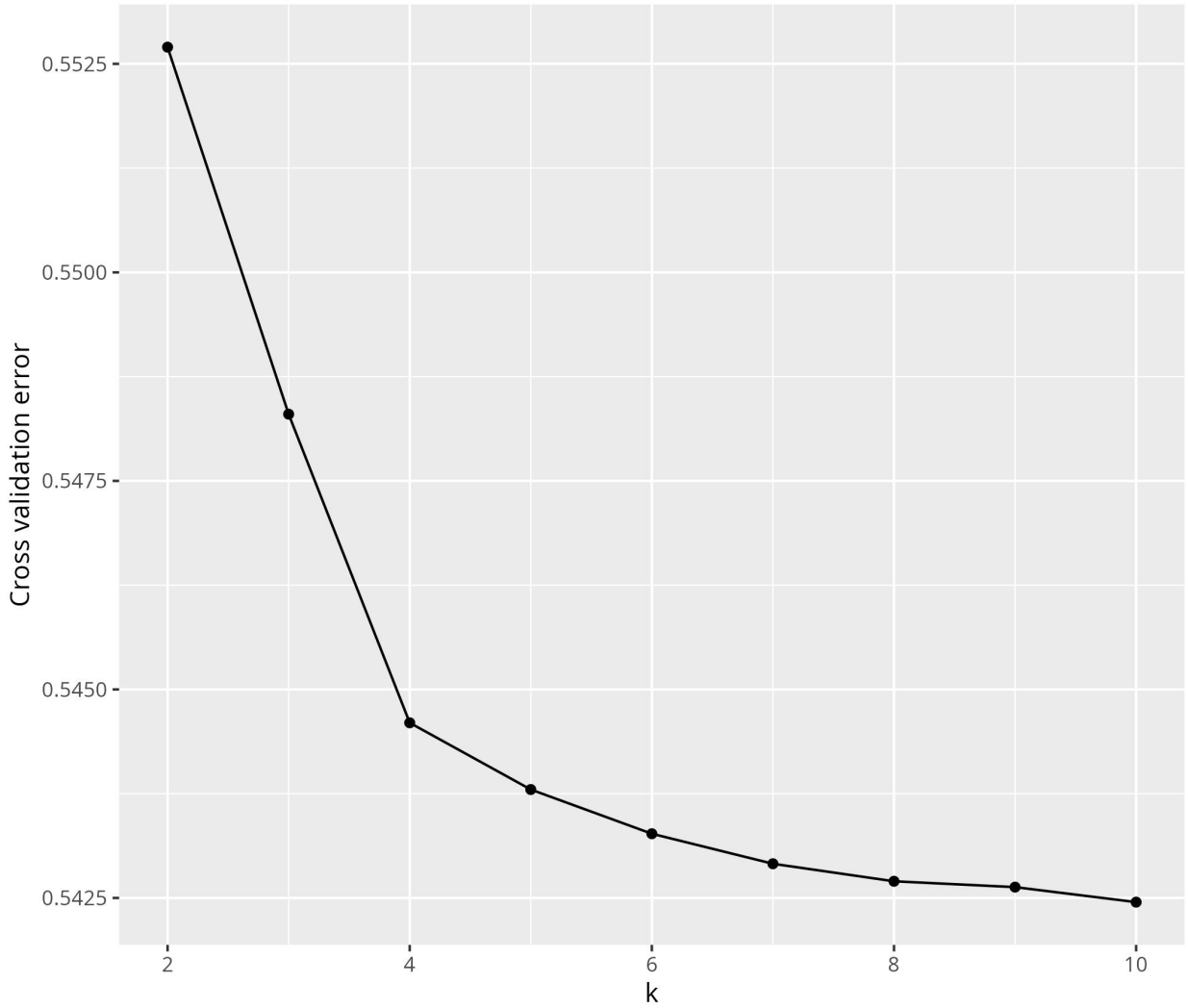


Figure 3: Supplementary Figure 3: Cross-validation error for admixture.

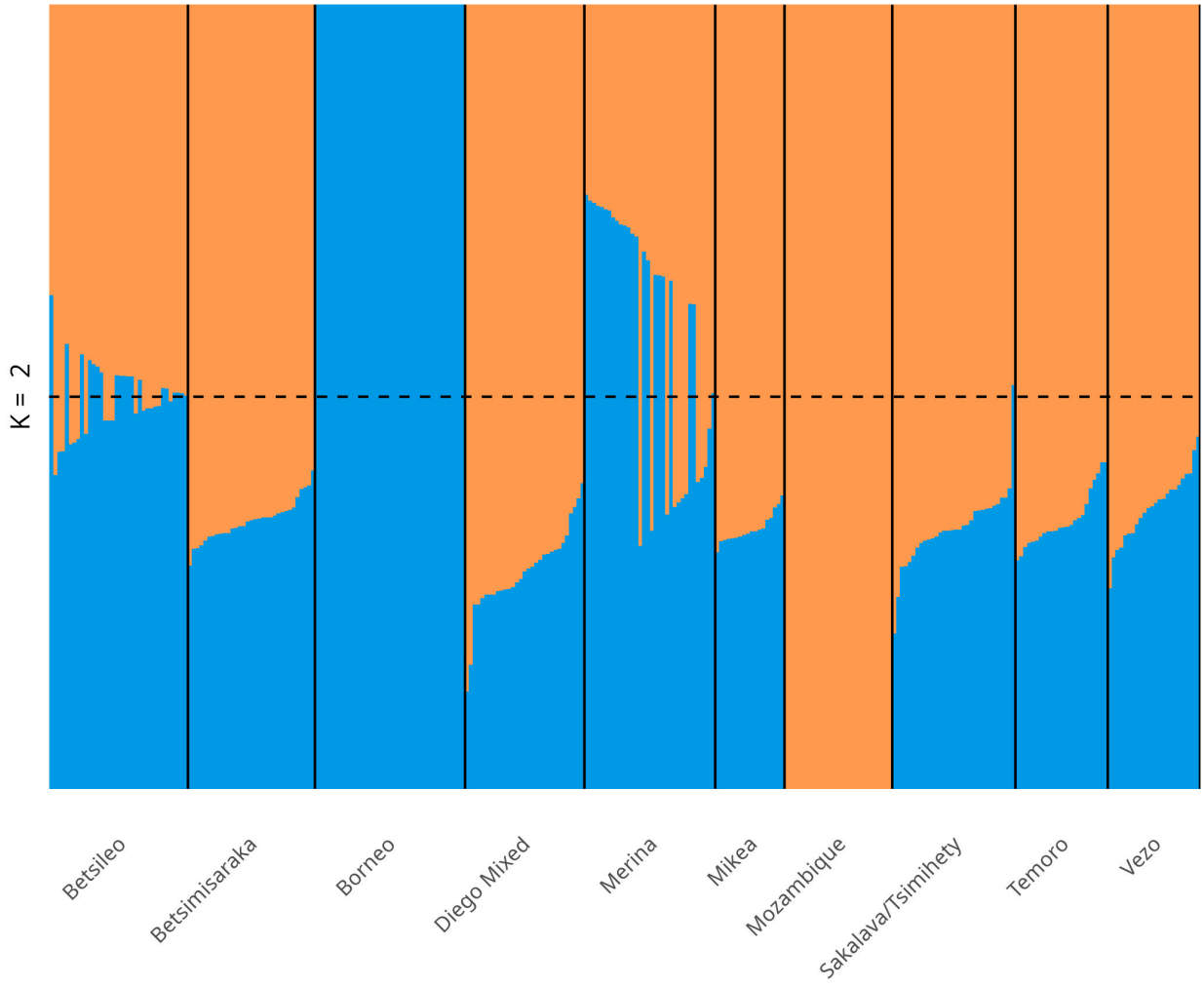


Figure 4: Supplementary Figure 4: Supervised admixture at K = 2 for Malagasy compared to reference populations. Individuals from Borneo from Brucato et al. (2016), and individuals from Mozambique populations from Fortes-Lima et al., (2024). Mozambique individuals are labelled together, but this group contains individuals given the following population labels: “Mozambique_Nhungwe”, “Mozambique_Tsonga”, “Mozambique_Ronga”, “Mozambique_TongaMoz”, “Mozambique_Chuwabu”, “Mozambique_Tswa”, “Mozambique_Chopi”, “Mozambique_Sena”, “Mozambique_Makhuwa”.

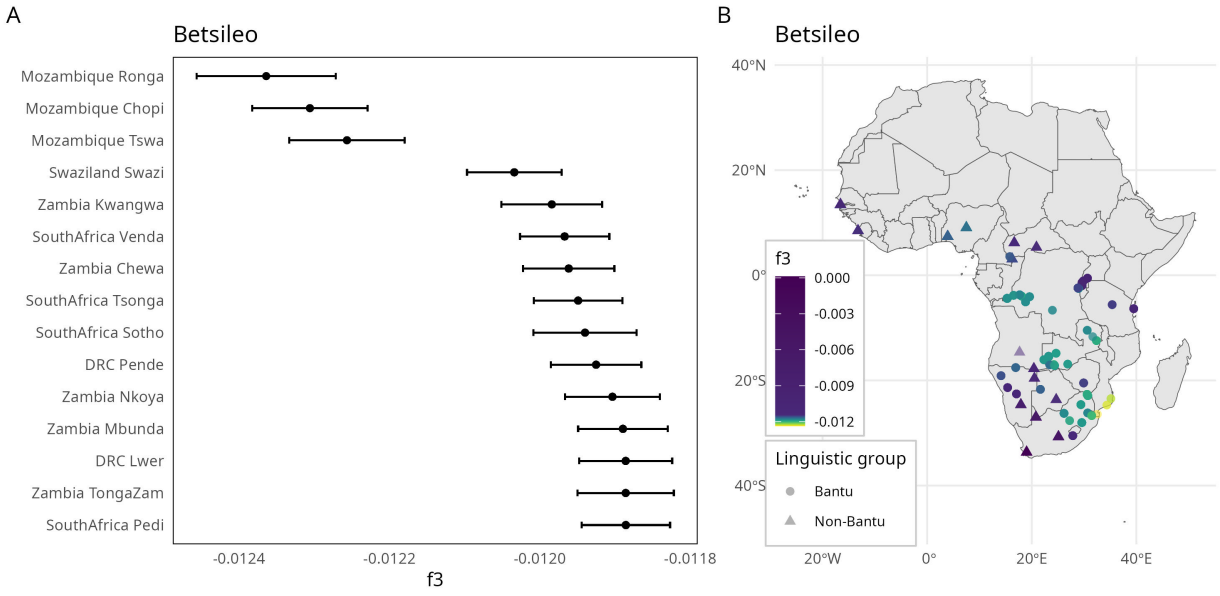


Figure 5: Supplementary Figure 5a: f3 statistic values across different African references for the Betsileo

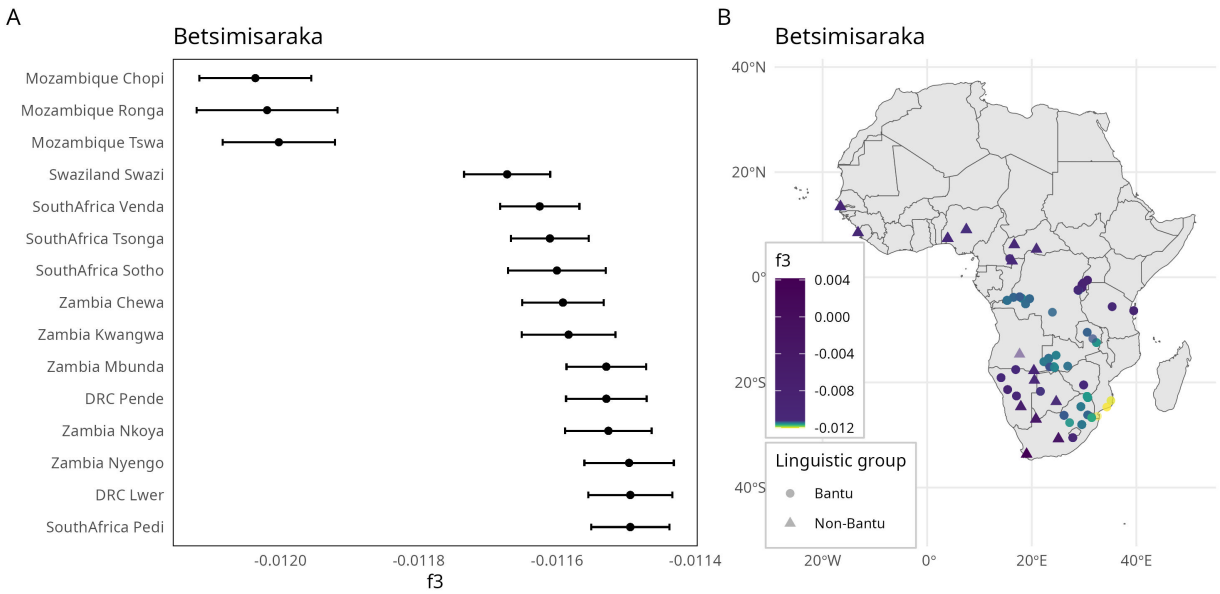


Figure 6: Supplementary Figure 5b: f3 statistic values across different African references for the Betsimisaraka

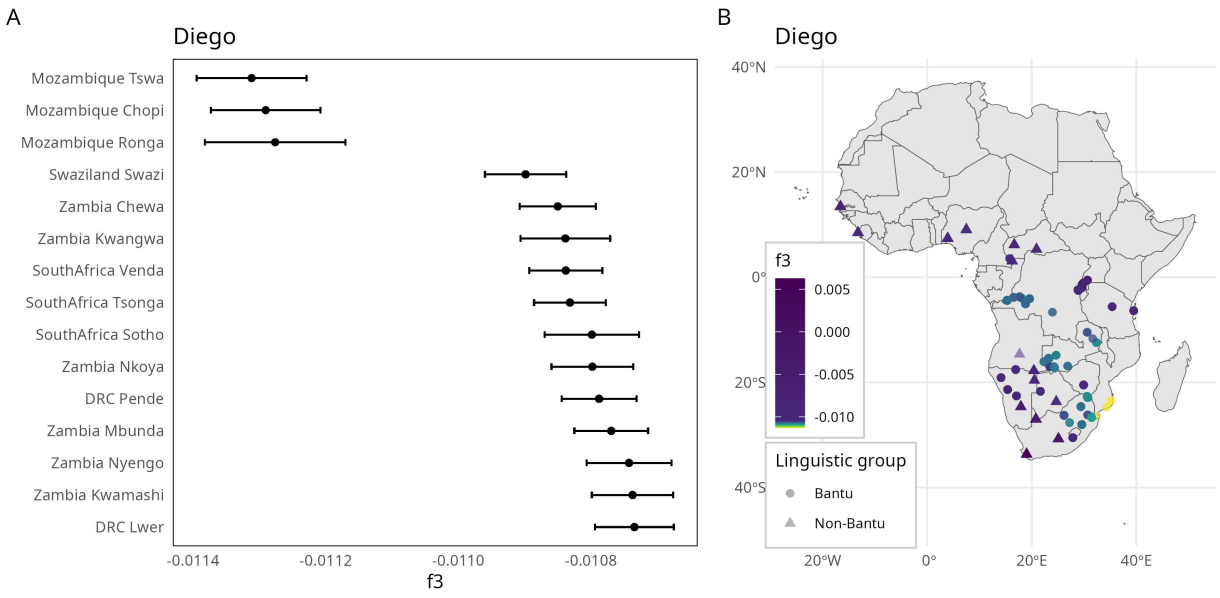


Figure 7: Supplementary Figure 5c: f3 statistic values across different African references for Diego

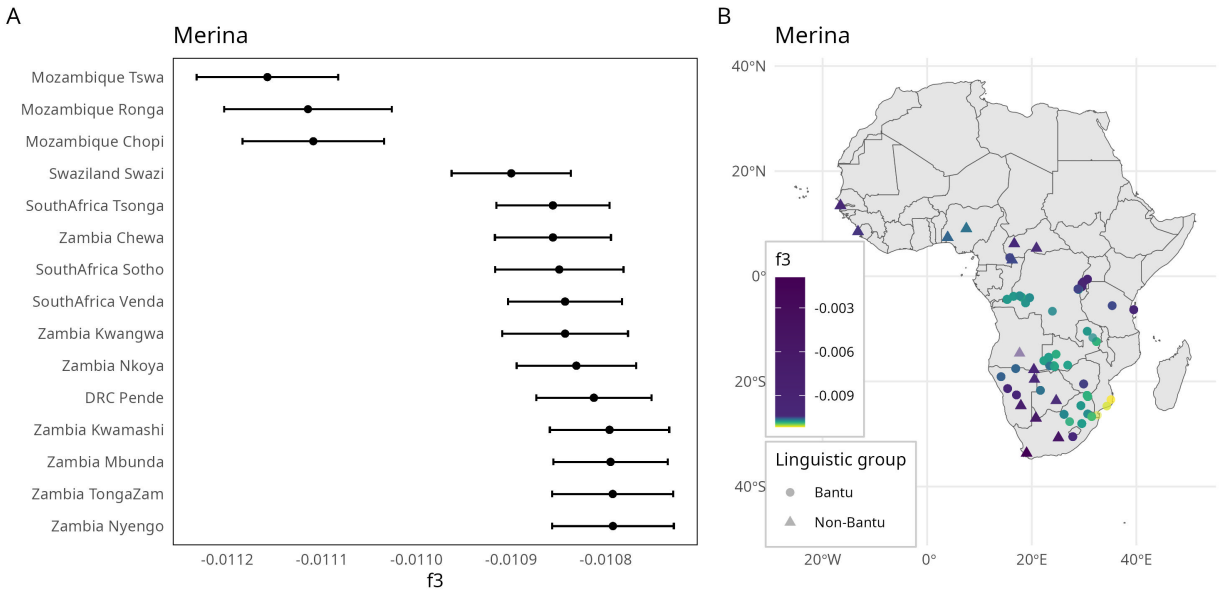


Figure 8: Supplementary Figure 5d: f3 statistic values across different African references for the Merina

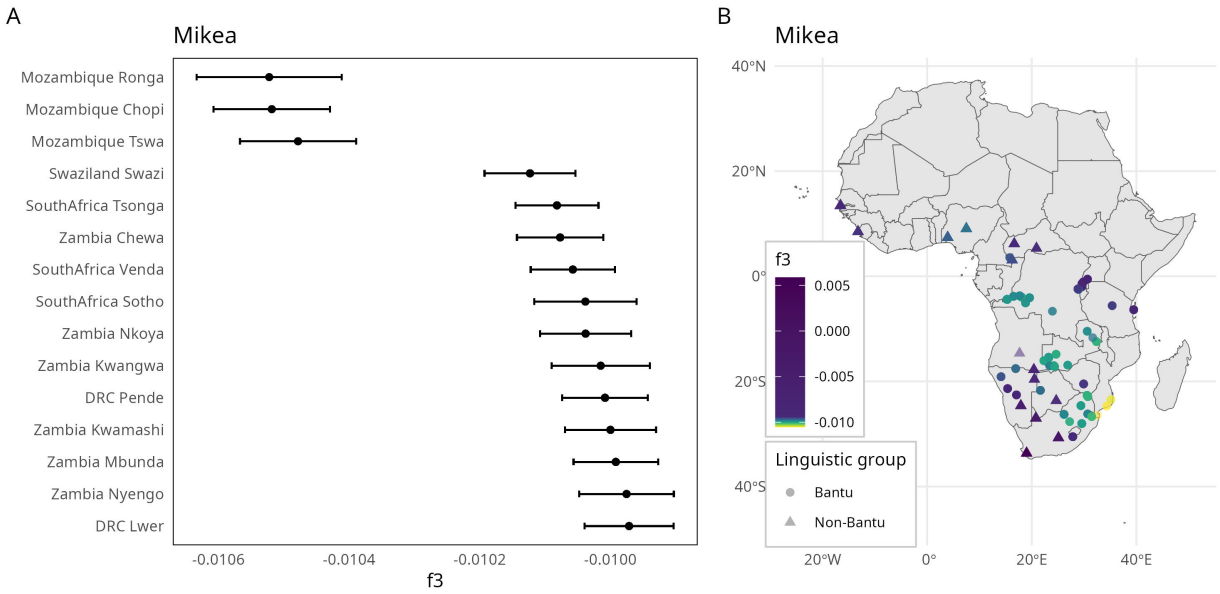


Figure 9: Supplementary Figure 5e: f3 statistic values across different African references for the Mikea

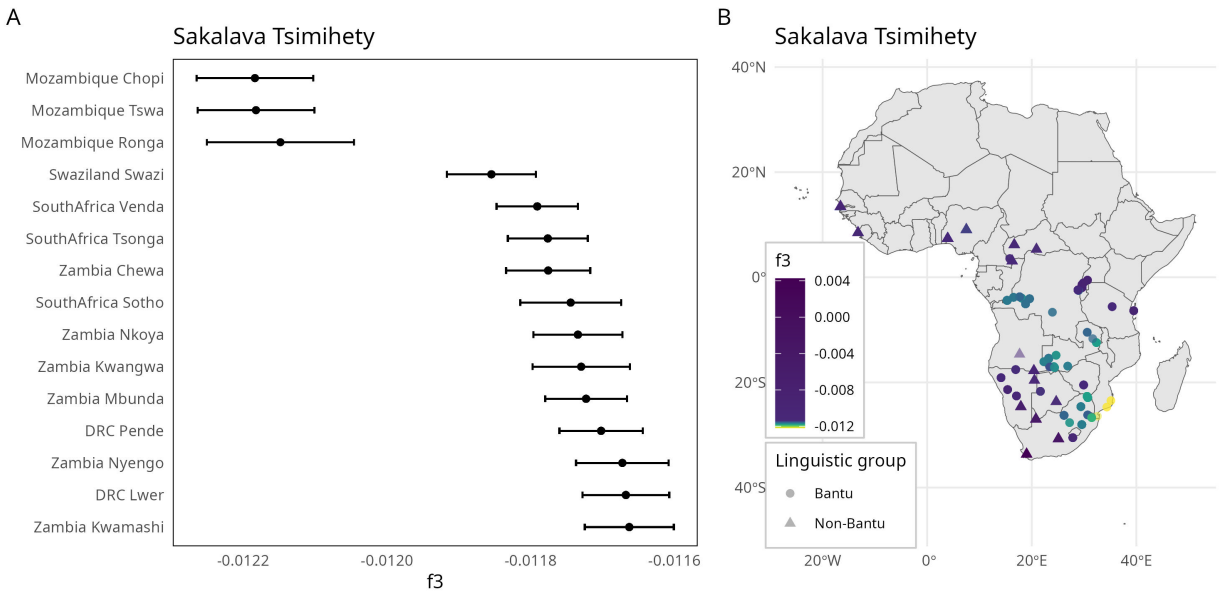


Figure 10: Supplementary Figure 5f: f3 statistic values across different African references for the Sakalava and Tsimihety

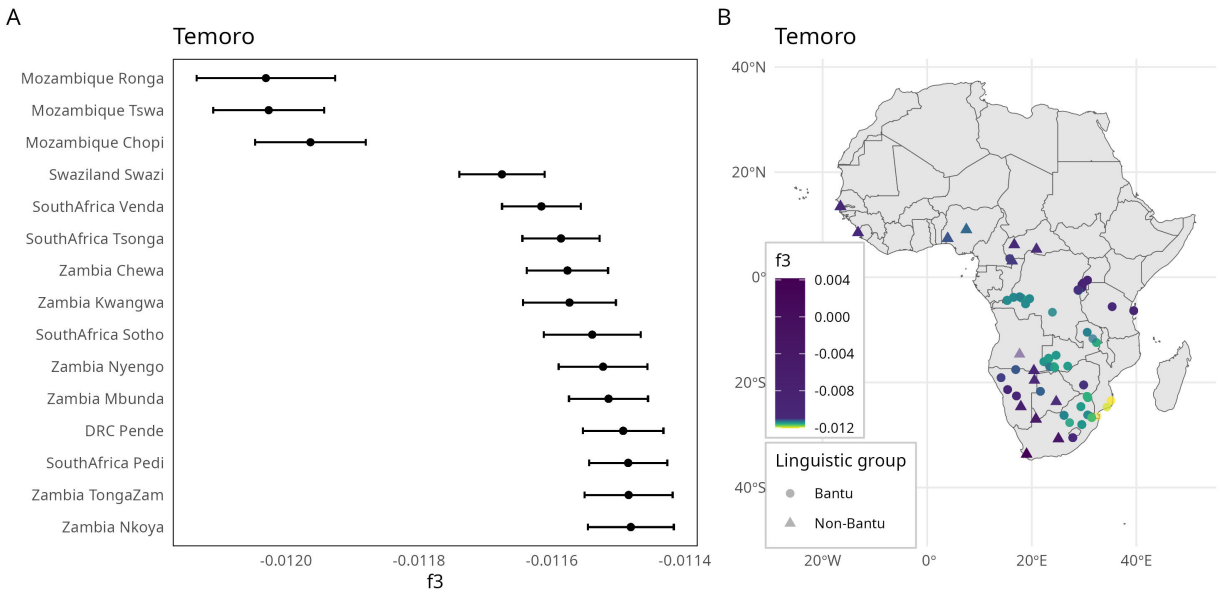


Figure 11: Supplementary Figure 5g: f3 statistic values across different African references for the Temoro

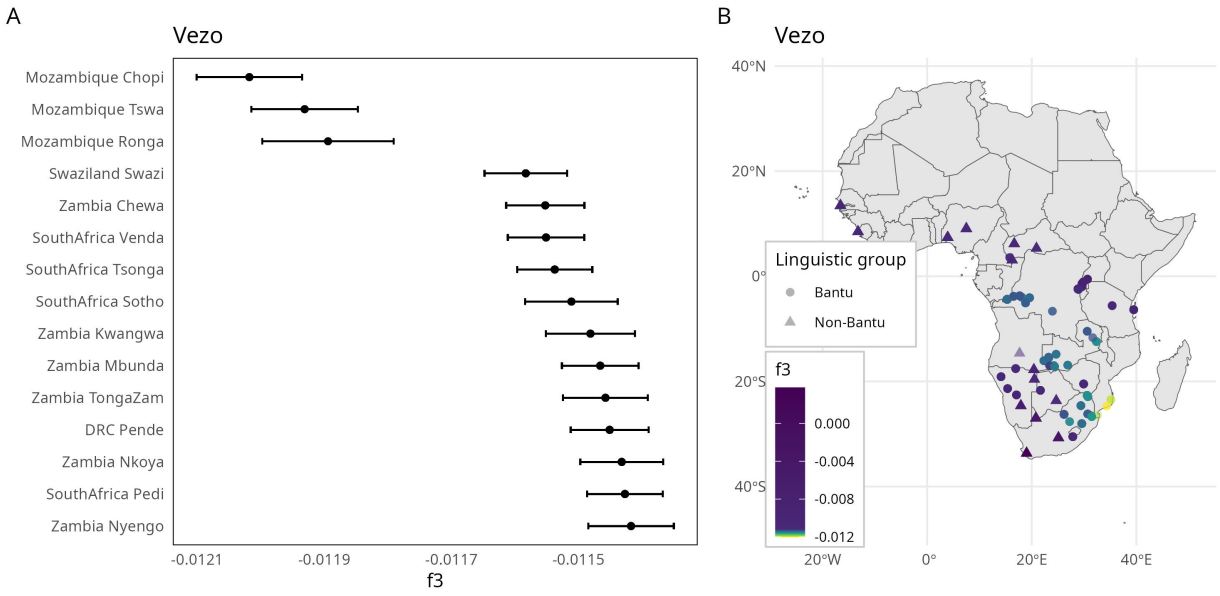


Figure 12: Supplementary Figure 5h: f3 statistic values across different African references for the Vezo

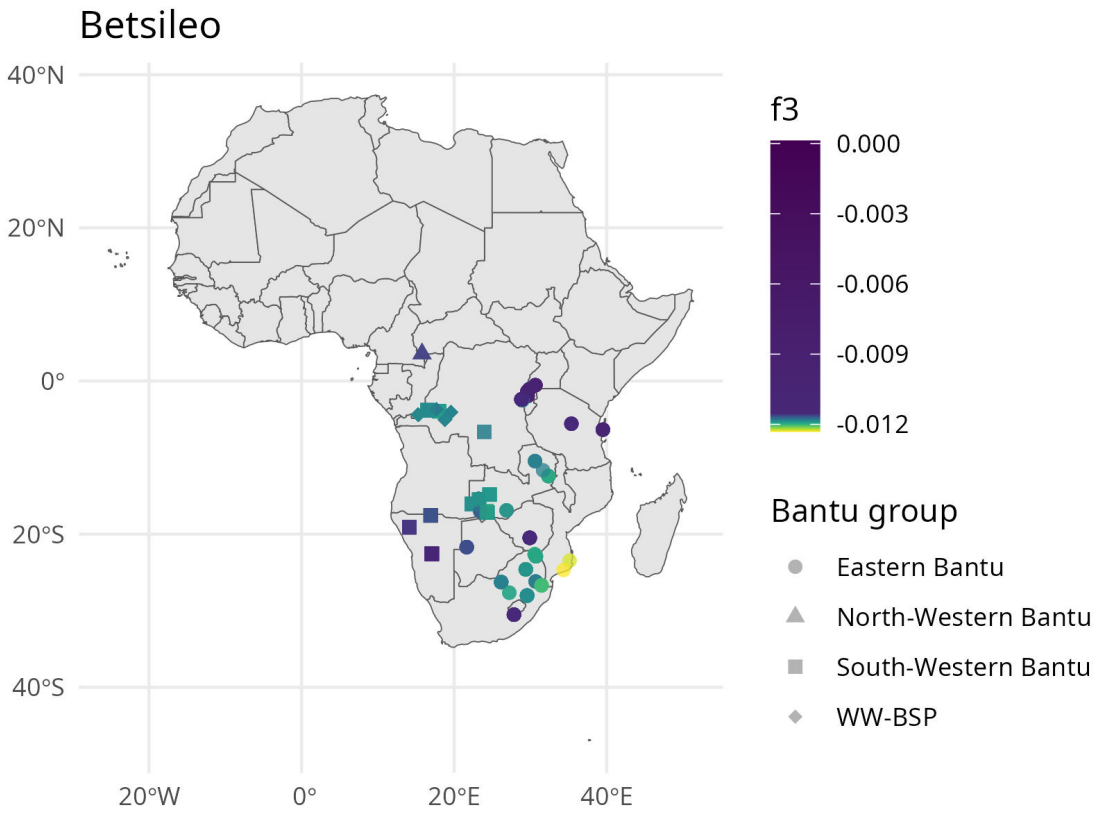


Figure 13: Supplementary Figure 6a: f3 statistic values across different Bantu African references for the Betsileo

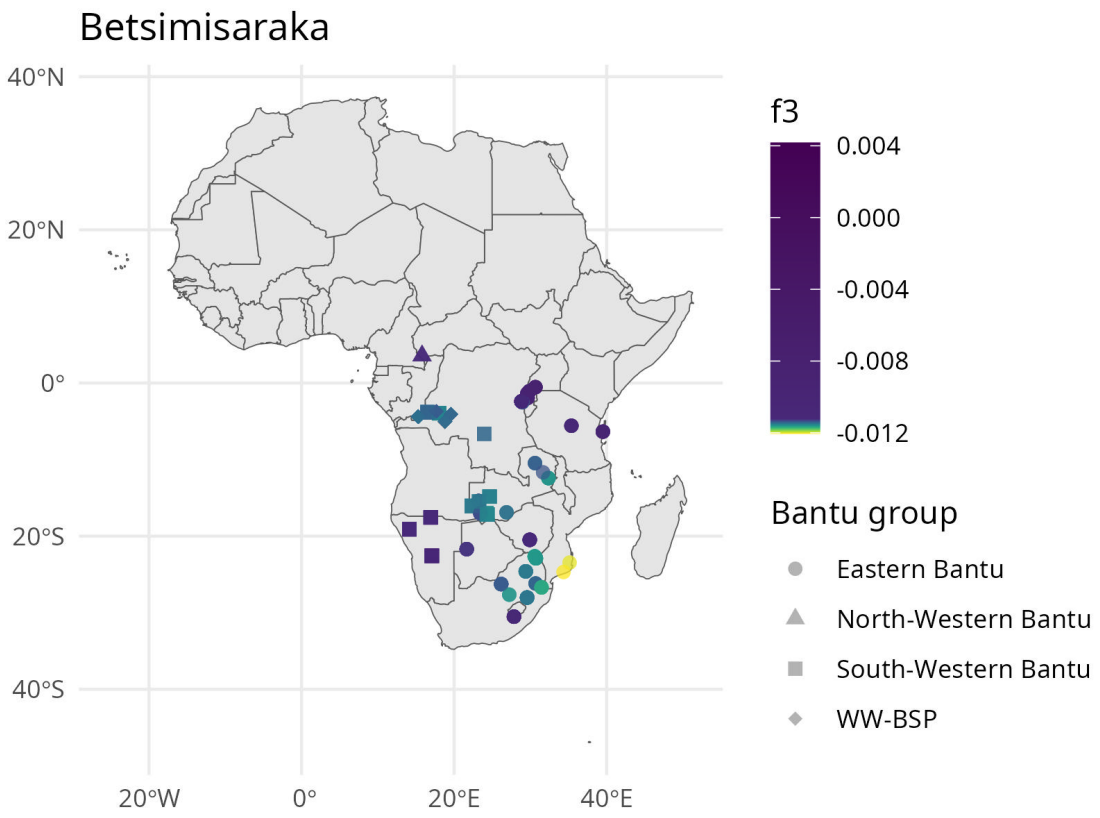


Figure 14: Supplementary Figure 6b: f3 statistic values across different Bantu African references for the Betsimisaraka

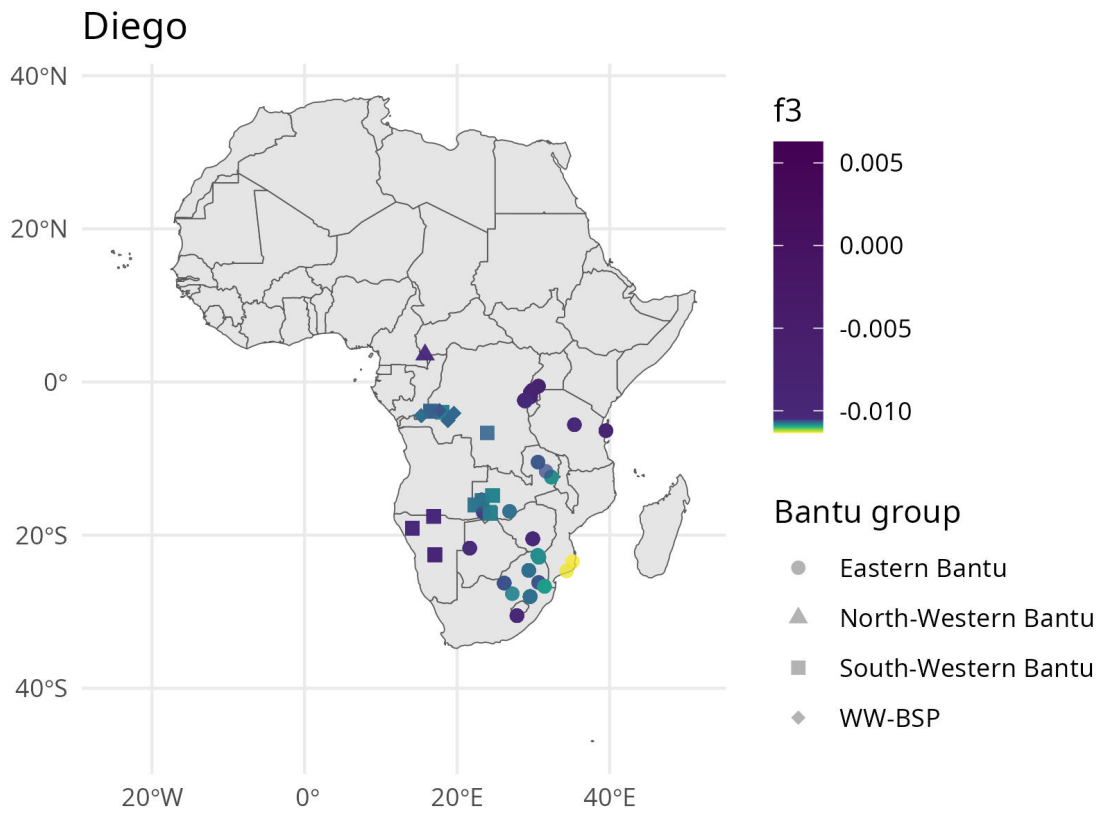


Figure 15: Supplementary Figure 6c: f3 statistic values across different Bantu African references for Diego

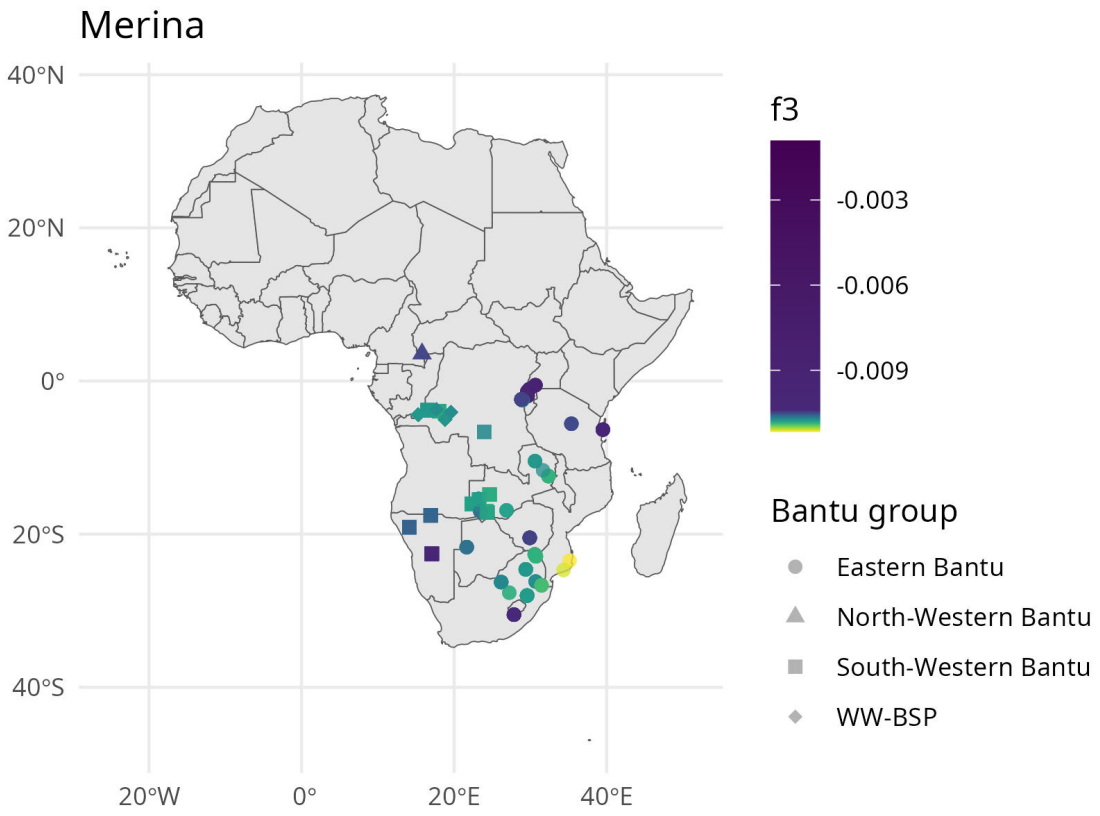


Figure 16: Supplementary Figure 6d: f3 statistic values across different Bantu African references for the Merina

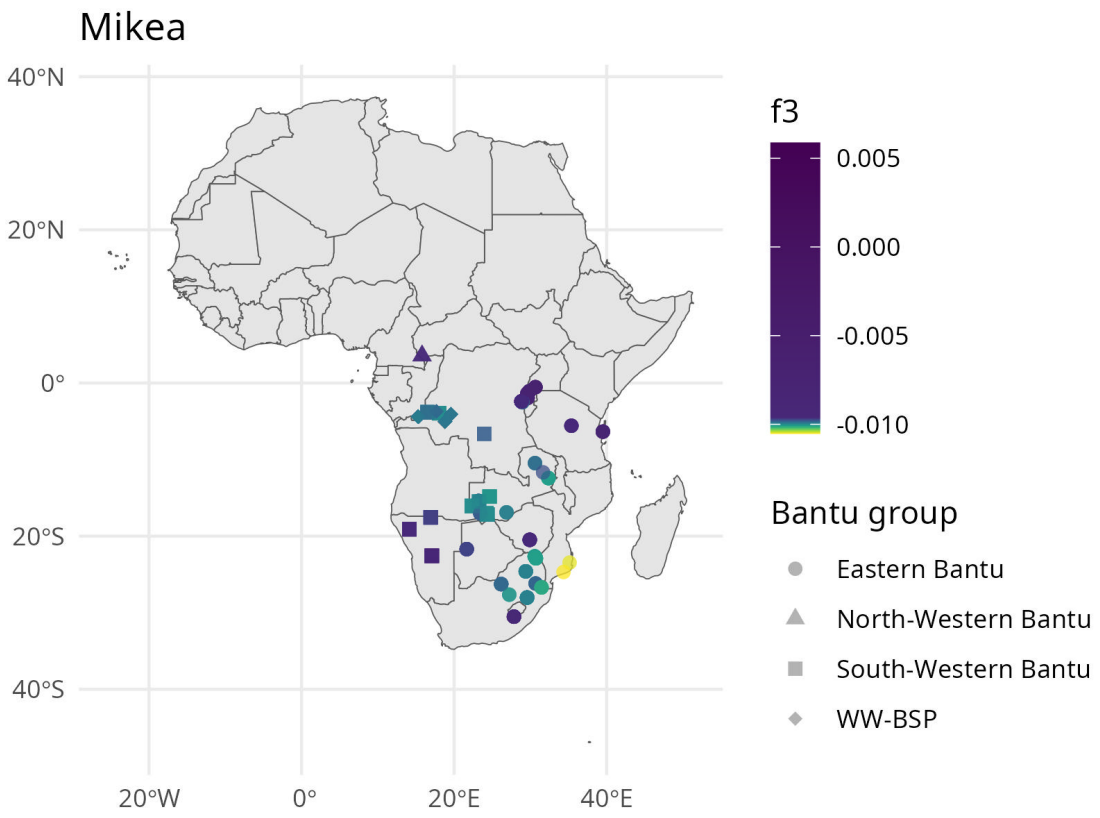


Figure 17: Supplementary Figure 6e: f3 statistic values across different Bantu African references for the Mikea

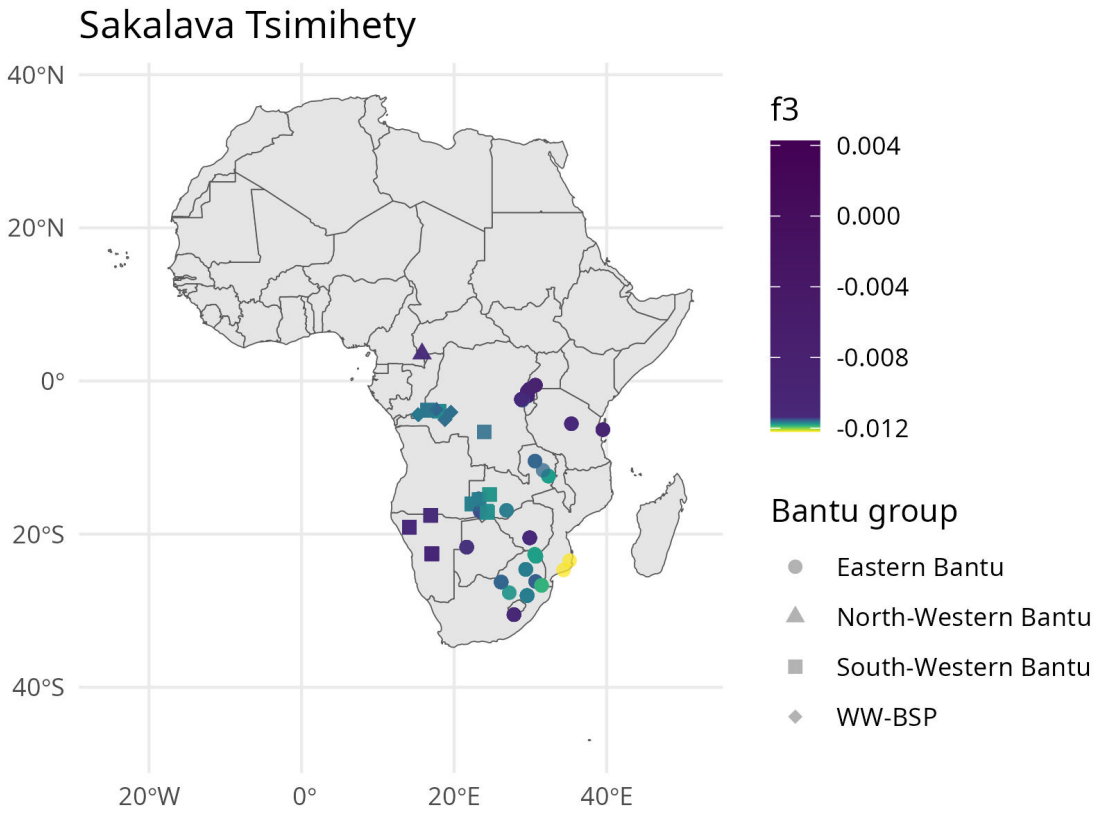


Figure 18: Supplementary Figure 6f: f3 statistic values across different Bantu African references for the Sakalava and Tsimihety

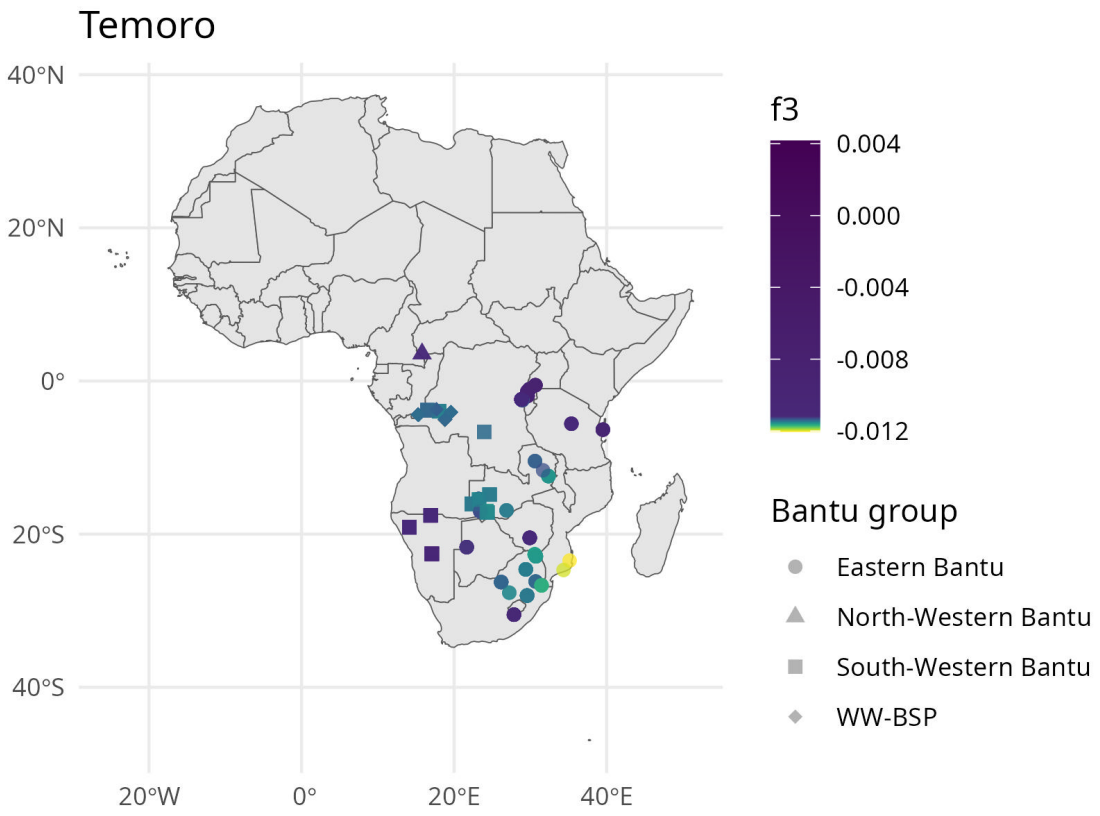


Figure 19: Supplementary Figure 6g: f3 statistic values across different Bantu African references for the Temoro

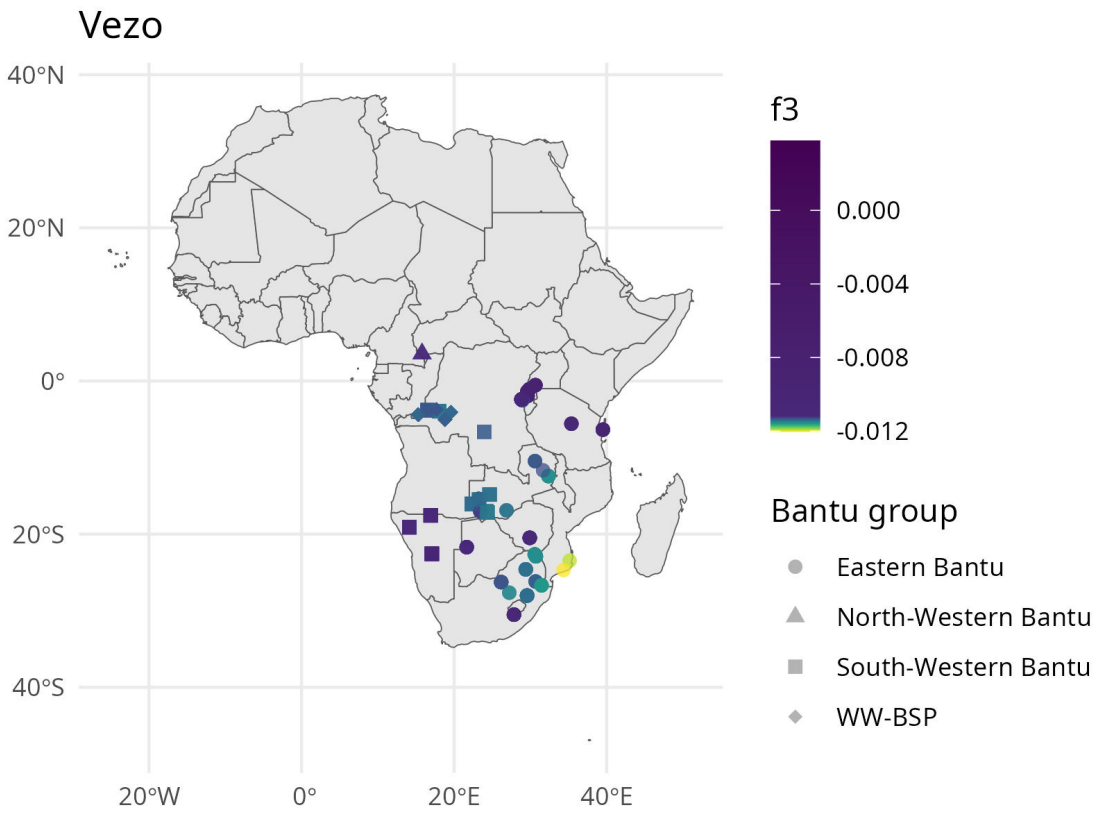


Figure 20: Supplementary Figure 6h: f3 statistic values across different Bantu African references for the Vezo

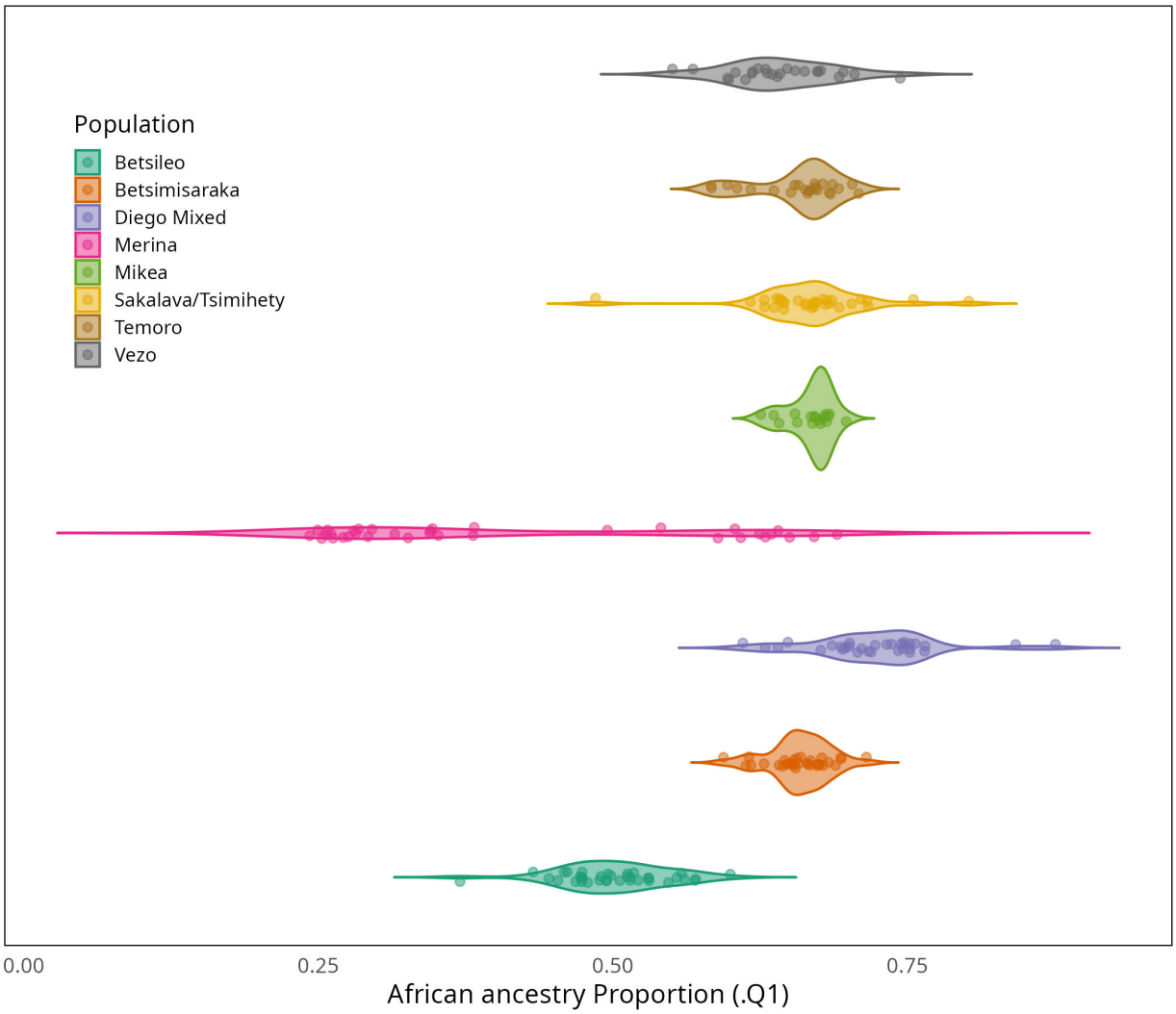


Figure 21: Supplementary Figure 7: African ancestry proportion for each Malagasy population derived from supervised admixture at K = 2.

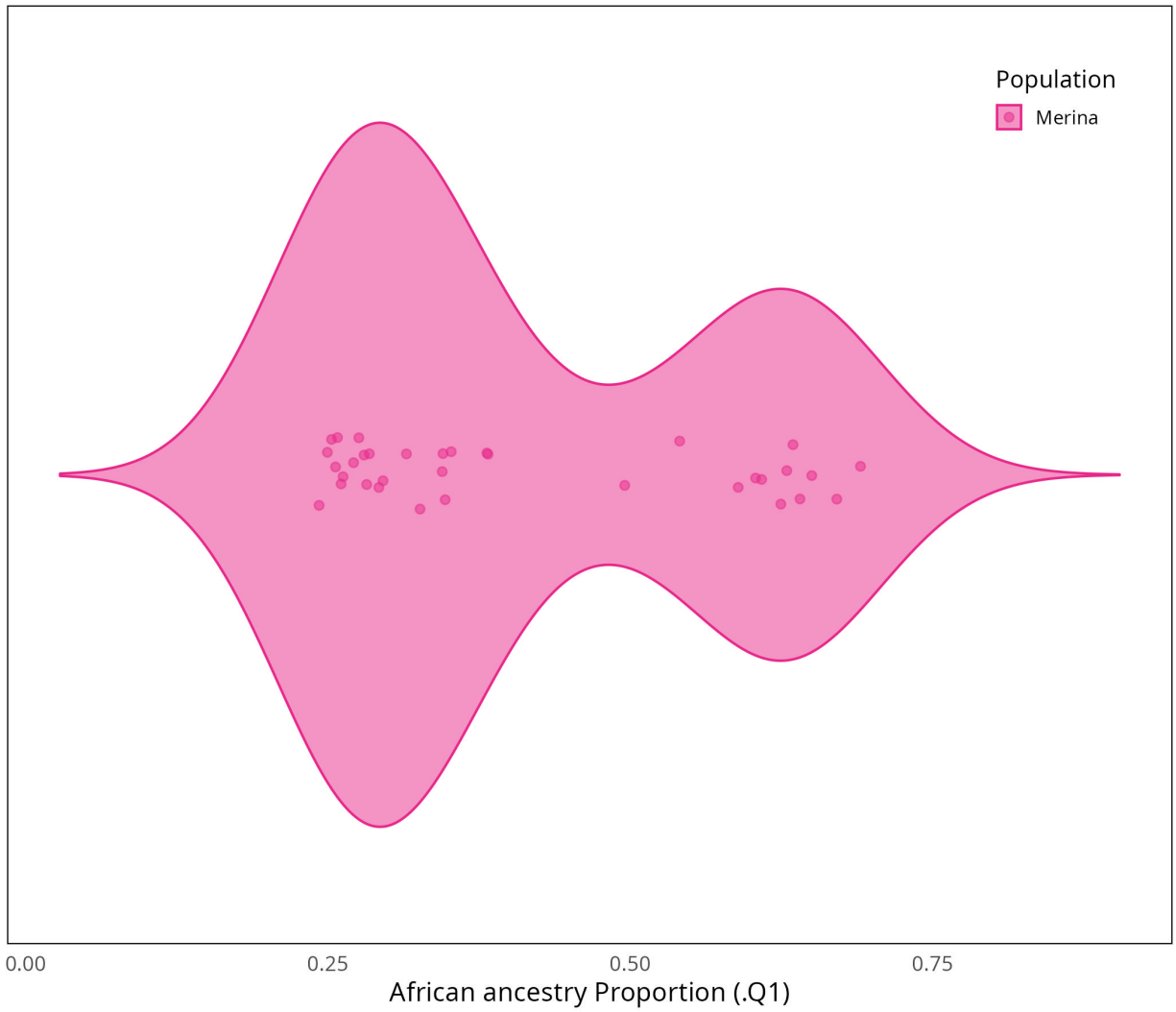


Figure 22: Supplementary Figure 8: African ancestry proportion for Merina individuals derived from supervised admixture at K = 2, showing the bimodal distribution of ancestry among Merina individuals..

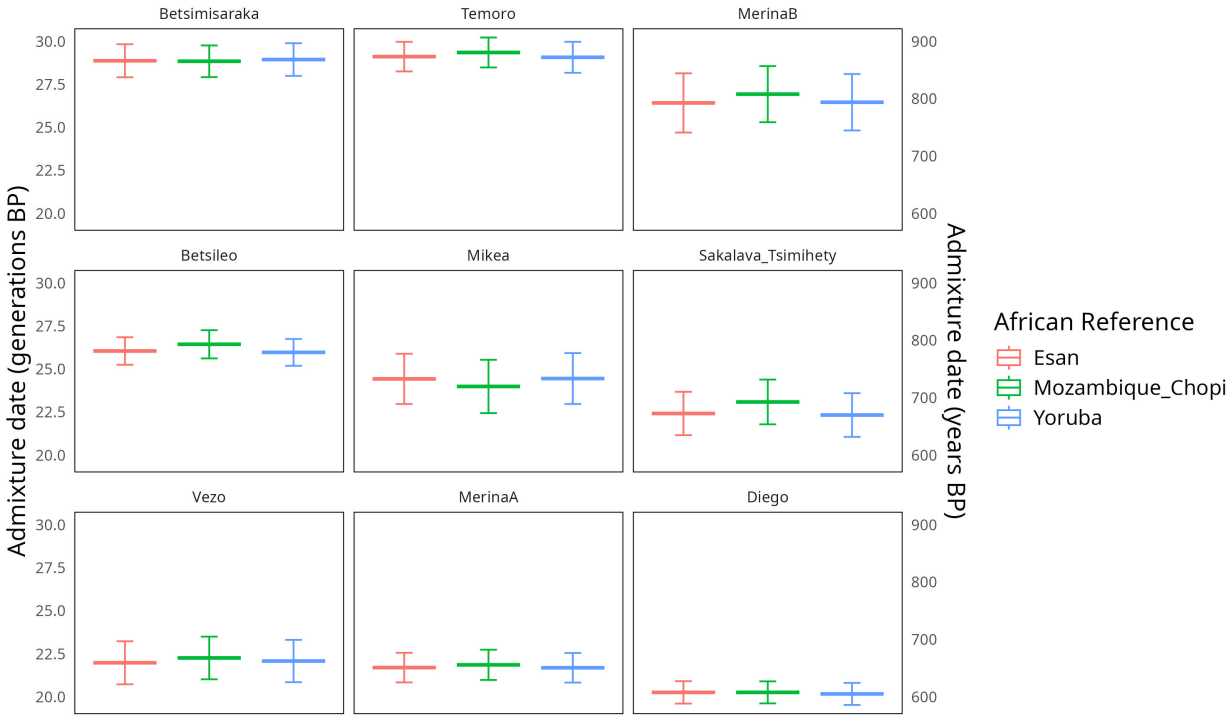


Figure 23: Supplementary Figure 9: Estimated admixture dates for Malagasy populations using MALDER (Pickrell et al., 2014), with Borneo as an Asian source population and varied African source populations. Horizontal bar indicates estimated date of admixture, vertical bars represent standard error from leave-one-chromosome-out jackknifing.

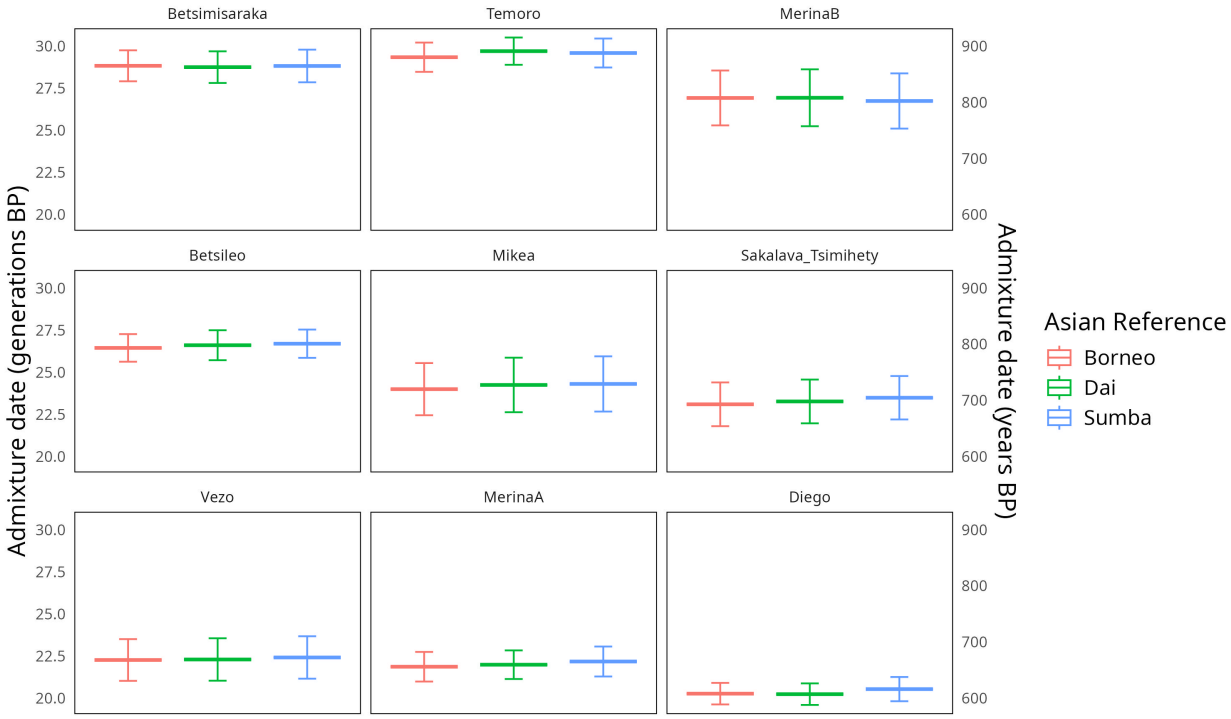


Figure 24: Supplementary Figure 10: Estimated admixture dates for Malagasy populations using MALDER (Pickrell et al., 2014), with Mozambique as an African source population and varied Asian source populations. Horizontal bar indicates estimated date of admixture, vertical bars represent standard error from leave-one-chromosome-out jackknifing.

Table 1: Sample sizes for each population in the global dataset after filtering.

Population	Sample size	source
Angola_Khwe	14	Fortes-Lima et al., 2024
Angola_Sekele	7	Fortes-Lima et al., 2024
Angola_Vasekele	4	Fortes-Lima et al., 2024
Angola_Xun	2	Fortes-Lima et al., 2024
Bengali, Bengal	86	1000 Genomes Project Consortium, 2015
Betsileo	36	This study
Betsimisaraka	33	This study
Borneo	39	Brucato et al., 2016
Botswana_Ghanzi	25	Fortes-Lima et al., 2024
Botswana_GuiGhanaKgal	9	Fortes-Lima et al., 2024
Botswana_Kalanga	3	Fortes-Lima et al., 2024
Botswana_Kgalagadi	6	Fortes-Lima et al., 2024
Botswana_TswanaBot	6	Fortes-Lima et al., 2024
CAR_Banda	10	Fortes-Lima et al., 2024
CAR_DzangaShangaPeople	16	Fortes-Lima et al., 2024
CAR_Gbaya	21	Fortes-Lima et al., 2024
CAR_Mpiemo	9	Fortes-Lima et al., 2024
CAR_Nzakara	8	Fortes-Lima et al., 2024
CAR_Zande	6	Fortes-Lima et al., 2024
Han, China	105	1000 Genomes Project Consortium, 2015
Diego	31	This study
DRC_Balobo	1	Fortes-Lima et al., 2024
DRC_Bandundu	1	Fortes-Lima et al., 2024
DRC_Bangubangu	1	Fortes-Lima et al., 2024
DRC_BanguBangu	2	Fortes-Lima et al., 2024
DRC_Binji	1	Fortes-Lima et al., 2024
DRC_Binza	1	Fortes-Lima et al., 2024
DRC_Boo	1	Fortes-Lima et al., 2024
DRC_Ciokwe	1	Fortes-Lima et al., 2024
DRC_Cokwe	1	Fortes-Lima et al., 2024
DRC_Ding	42	Fortes-Lima et al., 2024
DRC_Fang	1	Fortes-Lima et al., 2024
DRC_Haavu	1	Fortes-Lima et al., 2024
DRC_Hendo	1	Fortes-Lima et al., 2024
DRC_Hungan	3	Fortes-Lima et al., 2024
DRC_Kango	1	Fortes-Lima et al., 2024
DRC_Kanyoka	1	Fortes-Lima et al., 2024
DRC_Kuba	3	Fortes-Lima et al., 2024
DRC_Kusu	1	Fortes-Lima et al., 2024
DRC_Lele	5	Fortes-Lima et al., 2024
DRC_Lika	1	Fortes-Lima et al., 2024
DRC_Luba	1	Fortes-Lima et al., 2024
DRC_LubaKasai	2	Fortes-Lima et al., 2024
DRC_LubaKatanga	2	Fortes-Lima et al., 2024
DRC_LubaLulua	6	Fortes-Lima et al., 2024

Table 1: Sample sizes for each population in the global dataset after filtering. *(continued)*

Population	Sample size	source
DRC_Lulua	1	Fortes-Lima et al., 2024
DRC_Lunda	2	Fortes-Lima et al., 2024
DRC_Luozi	2	Fortes-Lima et al., 2024
DRC_Lwer	17	Fortes-Lima et al., 2024
DRC_Mangbetu	1	Fortes-Lima et al., 2024
DRC_Manyanga	35	Fortes-Lima et al., 2024
DRC_Mbaama	1	Fortes-Lima et al., 2024
DRC_Mbala	20	Fortes-Lima et al., 2024
DRC_Mbudza	2	Fortes-Lima et al., 2024
DRC_Mbunda	1	Fortes-Lima et al., 2024
DRC_Mbuun	31	Fortes-Lima et al., 2024
DRC_MbweneNgong	1	Fortes-Lima et al., 2024
DRC_Mongo	1	Fortes-Lima et al., 2024
DRC_MongoNkundo	8	Fortes-Lima et al., 2024
DRC_Mpe	4	Fortes-Lima et al., 2024
DRC_Mpiin	2	Fortes-Lima et al., 2024
DRC_Mpur	1	Fortes-Lima et al., 2024
DRC_Myene	1	Fortes-Lima et al., 2024
DRC_Nande	1	Fortes-Lima et al., 2024
DRC_Nandi	2	Fortes-Lima et al., 2024
DRC_Ndibu	7	Fortes-Lima et al., 2024
DRC_Ngbaka	1	Fortes-Lima et al., 2024
DRC_Ngbandi	1	Fortes-Lima et al., 2024
DRC_Ngombe	2	Fortes-Lima et al., 2024
DRC_Ngong	4	Fortes-Lima et al., 2024
DRC_Ngwi	67	Fortes-Lima et al., 2024
DRC_Nienge	1	Fortes-Lima et al., 2024
DRC_Nsambaan	1	Fortes-Lima et al., 2024
DRC_Nsong	2	Fortes-Lima et al., 2024
DRC_Ntandu	4	Fortes-Lima et al., 2024
DRC_NunuBobangi	3	Fortes-Lima et al., 2024
DRC_NunuMushie	1	Fortes-Lima et al., 2024
DRC_Nzadi	7	Fortes-Lima et al., 2024
DRC_Nzebi	1	Fortes-Lima et al., 2024
DRC_Pende	21	Fortes-Lima et al., 2024
DRC_Rega	12	Fortes-Lima et al., 2024
DRC_RundiHutu	1	Fortes-Lima et al., 2024
DRC_Ruund	1	Fortes-Lima et al., 2024
DRC_Sakata	3	Fortes-Lima et al., 2024
DRC_Sengele	1	Fortes-Lima et al., 2024
DRC_Shi	129	Fortes-Lima et al., 2024
DRC_Songe	3	Fortes-Lima et al., 2024
DRC_Teke	1	Fortes-Lima et al., 2024
DRC_Tetela	3	Fortes-Lima et al., 2024
DRC_Wonk	4	Fortes-Lima et al., 2024

Table 1: Sample sizes for each population in the global dataset after filtering. *(continued)*

Population	Sample size	source
DRC_Yaka	5	Fortes-Lima et al., 2024
DRC_Yans	27	Fortes-Lima et al., 2024
DRC_Yombe	9	Fortes-Lima et al., 2024
DRC_Zimba	1	Fortes-Lima et al., 2024
Esan, Nigeria	98	1000 Genomes Project Consortium, 2015
Mandinka, The Gambia	113	1000 Genomes Project Consortium, 2015
Kinh, Vietnam	90	1000 Genomes Project Consortium, 2015
Lesotho_SothoSouthern	7	Fortes-Lima et al., 2024
Luhye, Kenya	94	1000 Genomes Project Consortium, 2015
Merina	34	This study
Mikea	18	Pierron et al., 2014
Mozambique_Chopi	4	Fortes-Lima et al., 2024
Mozambique_Chuwabu	3	Fortes-Lima et al., 2024
Mozambique_Makhuwa	1	Fortes-Lima et al., 2024
Mozambique_Nhungwe	3	Fortes-Lima et al., 2024
Mozambique_Ronga	2	Fortes-Lima et al., 2024
Mozambique_Sena	1	Fortes-Lima et al., 2024
Mozambique_TongaMoz	3	Fortes-Lima et al., 2024
Mozambique_Tsonga	7	Fortes-Lima et al., 2024
Mozambique_Tswa	4	Fortes-Lima et al., 2024
Mende, Sierra Leone	84	1000 Genomes Project Consortium, 2015
Namibia_Damara	21	Fortes-Lima et al., 2024
Namibia_Herero	23	Fortes-Lima et al., 2024
Namibia_Himba	22	Fortes-Lima et al., 2024
Namibia_Kgalagadi	8	Fortes-Lima et al., 2024
Namibia_Nama	15	Fortes-Lima et al., 2024
Namibia_TsumkweKung	10	Fortes-Lima et al., 2024
Namibia_Wambo	22	Fortes-Lima et al., 2024
Punjabi, Pakistan	96	1000 Genomes Project Consortium, 2015
Rwanda_Nkore	18	Fortes-Lima et al., 2024
Sakalava_Tsimihety	32	This study
SouthAfrica_Bhaca	12	Fortes-Lima et al., 2024
SouthAfrica_Coloured	11	Fortes-Lima et al., 2024
SouthAfrica_ColouredColesberg	12	Fortes-Lima et al., 2024
SouthAfrica_ColouredWellington	19	Fortes-Lima et al., 2024
SouthAfrica_Hlubi	2	Fortes-Lima et al., 2024
SouthAfrica_Karretjie	9	Fortes-Lima et al., 2024
SouthAfrica_Khomani	28	Fortes-Lima et al., 2024
SouthAfrica_Mputhi	3	Fortes-Lima et al., 2024
SouthAfrica_Ndebele	8	Fortes-Lima et al., 2024
SouthAfrica_Pedi	28	Fortes-Lima et al., 2024
SouthAfrica_Shona	1	Fortes-Lima et al., 2024
SouthAfrica_Sotho	7	Fortes-Lima et al., 2024
SouthAfrica_SothoNorthern	20	Fortes-Lima et al., 2024
SouthAfrica_SothoSouthern	25	Fortes-Lima et al., 2024

Table 1: Sample sizes for each population in the global dataset after filtering. *(continued)*

Population	Sample size	source
SouthAfrica_Soto	1	Fortes-Lima et al., 2024
SouthAfrica_Swazi	5	Fortes-Lima et al., 2024
SouthAfrica_Tsonga	26	Fortes-Lima et al., 2024
SouthAfrica_Tswana	32	Fortes-Lima et al., 2024
SouthAfrica_Venda	22	Fortes-Lima et al., 2024
SouthAfrica_Xhosa	30	Fortes-Lima et al., 2024
SouthAfrica_Zulu	52	Fortes-Lima et al., 2024
Sumba	29	Cox et al., 2016
Swaziland_Swazi	14	Fortes-Lima et al., 2024
Tanzania_Mixed	1	Fortes-Lima et al., 2024
Tanzania_TanzaniaMixed	14	Fortes-Lima et al., 2024
Temoro	24	Pierron et al., 2014
Timor	7	Cox et al., 2016
Uganda_Fumbira	15	Fortes-Lima et al., 2024
Uganda_Kiga	15	Fortes-Lima et al., 2024
Uganda_Konzo	5	Fortes-Lima et al., 2024
Uganda_Nkore	18	Fortes-Lima et al., 2024
Vezo	24	Pierron et al., 2014
Yoruba, Nigeria	104	1000 Genomes Project Consortium, 2015
Zambia_Bemba	11	Fortes-Lima et al., 2024
Zambia_Chewa	17	Fortes-Lima et al., 2024
Zambia_Fwe	14	Fortes-Lima et al., 2024
Zambia_Kalanga	12	Fortes-Lima et al., 2024
Zambia_Kaonde	4	Fortes-Lima et al., 2024
Zambia_Kavango	1	Fortes-Lima et al., 2024
Zambia_Kgalagadi	10	Fortes-Lima et al., 2024
Zambia_Kwamashi	16	Fortes-Lima et al., 2024
Zambia_Kwangari	1	Fortes-Lima et al., 2024
Zambia_Kwangwa	8	Fortes-Lima et al., 2024
Zambia_Lozi	23	Fortes-Lima et al., 2024
Zambia_Luyana	5	Fortes-Lima et al., 2024
Zambia_Mbukushu	8	Fortes-Lima et al., 2024
Zambia_Mbunda	25	Fortes-Lima et al., 2024
Zambia_Ngoni	7	Fortes-Lima et al., 2024
Zambia_Nkoya	14	Fortes-Lima et al., 2024
Zambia_Nsenga	4	Fortes-Lima et al., 2024
Zambia_Nyengo	10	Fortes-Lima et al., 2024
Zambia_Shanjo	9	Fortes-Lima et al., 2024
Zambia_Shona	1	Fortes-Lima et al., 2024
Zambia_ShonaZam	4	Fortes-Lima et al., 2024
Zambia_TongaZam	10	Fortes-Lima et al., 2024
Zambia_Tswana	2	Fortes-Lima et al., 2024
Zambia_Tumbuka	4	Fortes-Lima et al., 2024
Zanzibar_Swahili	22	Fortes-Lima et al., 2024
Zimbabwe_Remba	32	Fortes-Lima et al., 2024

Table 1: Sample sizes for each population in the global dataset after filtering. *(continued)*

Population	Sample size	source
Total	2744	Total

Table 2: Summary of analyses and software used in this study.

Analysis	Package/function	Dataset	Individuals Removed	n indivs	Loci filtering	n SNPs
Malagasy PCA	tidypopgen: gt_pca_randomSVD	Malagasy dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	233	227410
Global PCA	tidypopgen: gt_pca_randomSVD	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2744	198196
Unsupervised clustering	tidypopgen: gt_admixture	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2744	198196
Supervised clustering	ADMIXTURE: ADMIXTURE	Subset of Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	299	198196
F3 statistics	tidypopgen: gt_extract_f2; admixtools2 (version 2.0.8) (Maier and Patterson, 2024): qp3pop	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2744	198196
F4 ratio statistics	tidypopgen: gt_extract_f2; admixtools2 (version 2.0.8) (Maier and Patterson, 2024): qpf4ratio	Global dataset + Pan and European outgroups	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2792	77993
MALDER	MALDER (version 1.0) (Pickrell et al., 2014): MALDER	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025	2744	425904
Pairwise FST	tidypopgen: pairwise_pop_fst - method = "WC84" (Weir and Cockerham, 1984)	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2744	198196

Table 3: Minimum f3 statistic values across all combinations of references for each Malagasy population.

Malagasy	Asian	African	est	se	z	p
Betsileo	Borneo	Mozambique_Ronga	-0.0123638	0.0000925	-133.67820	0
Betsimisaraka	Borneo	Mozambique_Chopi	-0.0120364	0.0000810	-148.58346	0
Diego	Borneo	Mozambique_Tswa	-0.0113118	0.0000828	-136.57844	0
Merina	Borneo	Mozambique_Tswa	-0.0111576	0.0000747	-149.26685	0
Mikea	Borneo	Mozambique_Ronga	-0.0105223	0.0001111	-94.68672	0
Sakalava Tsimihety	Borneo	Mozambique_Chopi	-0.0121836	0.0000811	-150.27787	0
Temoro	Borneo	Mozambique_Ronga	-0.0120305	0.0001042	-115.42556	0
Vezo	Borneo	Mozambique_Chopi	-0.0120179	0.0000829	-144.89537	0

Table 4: Minimum z-scores for f3 statistics across all combinations for each Malagasy population.

Malagasy	Asian	African	est	se	z	p
Betsileo	Borneo	DRC_Shi	-0.0115276	5.51e-05	-209.2158	0
Betsimisaraka	Borneo	DRC_Shi	-0.0110031	5.16e-05	-213.0884	0
Diego	Borneo	DRC_Shi	-0.0102260	4.85e-05	-210.7807	0
Merina	Borneo	DRC_Shi	-0.0104804	5.67e-05	-184.9265	0
Mikea	Borneo	DRC_Ngwi	-0.0099074	6.14e-05	-161.3010	0
Sakalava Tsimihety	Borneo	DRC_Shi	-0.0111734	5.08e-05	-220.1194	0
Temoro	Borneo	DRC_Shi	-0.0110101	5.37e-05	-205.2034	0
Vezo	Borneo	DRC_Shi	-0.0109646	5.46e-05	-200.6620	0

Table 5: f4 ratios calculated with admixtools. f4 ratios were calculated in the configurations f4(GBR,Chimpanzee;Malagasy,Borneo) / f4(GBR,Chimpanzee;Esan,Borneo) and f4(GBR,Chimpanzee;Malagasy,Esan) / f4(GBR,Chimpanzee;Borneo,Esan)

pop1	pop2	pop3	pop4	pop5	alpha	se	z
GBR	chimp_ref	Betsimisaraka	Borneo	ESN	0.5946328	0.0090260	65.87983
GBR	chimp_ref	MerinaA	Borneo	ESN	0.2431837	0.0120343	20.20760
GBR	chimp_ref	MerinaB	Borneo	ESN	0.5501539	0.0133817	41.11243
GBR	chimp_ref	Diego	Borneo	ESN	0.6109459	0.0090605	67.42992
GBR	chimp_ref	Sakalava_Tsimihety	Borneo	ESN	0.5839059	0.0089299	65.38787
GBR	chimp_ref	Betsileo	Borneo	ESN	0.4564963	0.0095466	47.81746
GBR	chimp_ref	Temoro	Borneo	ESN	0.5708479	0.0099019	57.65021
GBR	chimp_ref	Vezo	Borneo	ESN	0.5554886	0.0101537	54.70806
GBR	chimp_ref	Mikea	Borneo	ESN	0.5831392	0.0115808	50.35376

Table 6: MALDER results for all Malagasy populations, with varied African and Asian source populations. p value: significance of model tested. test pop: Malagasy target population. ref A: Asian population. ref B: African population. 2 ref z score: z score. 2 ref decay: Date, in generations before present, since admixture began with standard error. 2 ref amp exp: Amplitude of the LD decay curve. amp aff: Affine term estimating population substructure.

p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp	amp_aff
4.6e-222	Betsileo	Borneo	Esan	31.81	26.05 +/- 0.80	0.00216033 +/- 0.00006791	0.0000299
3.6e-234	Betsileo	Borneo	Yoruba	32.67	25.97 +/- 0.78	0.00214609 +/- 0.00006568	0.0000296
5.7e-230	Betsileo	Borneo	Mozambique_Chopi	32.38	26.44 +/- 0.82	0.00227784 +/- 0.00006829	0.0000310
3.5e-188	Betsileo	Sumba	Esan	29.26	26.26 +/- 0.82	0.00191612 +/- 0.00006549	0.0000261
1.7e-196	Betsileo	Sumba	Yoruba	29.91	26.18 +/- 0.80	0.00190193 +/- 0.00006360	0.0000258
1.5e-200	Betsileo	Sumba	Mozambique_Chopi	30.22	26.69 +/- 0.84	0.00202856 +/- 0.00006714	0.0000271
8.6e-200	Betsileo	Dai	Esan	30.16	26.21 +/- 0.87	0.00216652 +/- 0.00006781	0.0000304
1.5e-205	Betsileo	Dai	Yoruba	30.59	26.14 +/- 0.85	0.00215206 +/- 0.00006551	0.0000300
1.3e-197	Betsileo	Dai	Mozambique_Chopi	29.99	26.60 +/- 0.89	0.00228532 +/- 0.00006899	0.0000315
1.1e-196	Betsimisaraka	Borneo	Esan	29.92	28.87 +/- 0.96	0.00187297 +/- 0.00004607	0.0000083
7.2e-204	Betsimisaraka	Borneo	Yoruba	30.47	28.94 +/- 0.95	0.00186972 +/- 0.00004541	0.0000081
6.9e-217	Betsimisaraka	Borneo	Mozambique_Chopi	31.43	28.84 +/- 0.92	0.00197274 +/- 0.00004997	0.0000086
4.5e-185	Betsimisaraka	Sumba	Esan	29.01	28.90 +/- 1.00	0.00165692 +/- 0.00005122	0.0000072
2.2e-189	Betsimisaraka	Sumba	Yoruba	29.35	28.96 +/- 0.99	0.00165296 +/- 0.00005075	0.0000070
1.6e-193	Betsimisaraka	Sumba	Mozambique_Chopi	29.68	28.83 +/- 0.97	0.00175133 +/- 0.00005520	0.0000074
1.3e-199	Betsimisaraka	Dai	Esan	30.14	28.81 +/- 0.96	0.00186998 +/- 0.00004787	0.0000080
6.6e-205	Betsimisaraka	Dai	Yoruba	30.54	28.88 +/- 0.95	0.00186641 +/- 0.00004697	0.0000078
3.4e-204	Betsimisaraka	Dai	Mozambique_Chopi	30.49	28.76 +/- 0.94	0.00196995 +/- 0.00005151	0.0000083
1.5e-144	Diego	Borneo	Esan	25.60	20.26 +/- 0.65	0.00146813 +/- 0.00005735	0.0000493
4.1e-148	Diego	Borneo	Yoruba	25.92	20.17 +/- 0.64	0.00145881 +/- 0.00005628	0.0000494
4.4e-159	Diego	Borneo	Mozambique_Chopi	26.87	20.26 +/- 0.64	0.00153534 +/- 0.00005713	0.0000514
2.2e-116	Diego	Sumba	Esan	22.93	20.57 +/- 0.73	0.00131657 +/- 0.00005741	0.0000432
9.8e-120	Diego	Sumba	Yoruba	23.27	20.46 +/- 0.71	0.00130741 +/- 0.00005619	0.0000433
6.7e-124	Diego	Sumba	Mozambique_Chopi	23.67	20.53 +/- 0.72	0.00137919 +/- 0.00005826	0.0000452
4.7e-130	Diego	Dai	Esan	24.26	20.24 +/- 0.66	0.00146933 +/- 0.00006055	0.0000496
1.1e-133	Diego	Dai	Yoruba	24.60	20.14 +/- 0.64	0.00145999 +/- 0.00005934	0.0000497
2.3e-138	Diego	Dai	Mozambique_Chopi	25.04	20.23 +/- 0.64	0.00153750 +/- 0.00006140	0.0000517
1.2e-228	Merina	Borneo	Esan	32.28	22.65 +/- 0.70	0.00176093 +/- 0.00005003	0.0004215
5.5e-236	Merina	Borneo	Yoruba	32.80	22.66 +/- 0.69	0.00175499 +/- 0.00004855	0.0004206
1.5e-226	Merina	Borneo	Mozambique_Chopi	32.13	23.18 +/- 0.72	0.00186187 +/- 0.00004419	0.0004401
1.1e-227	Merina	Sumba	Esan	32.21	22.68 +/- 0.65	0.00154560 +/- 0.00004798	0.0003678
2.2e-238	Merina	Sumba	Yoruba	32.97	22.68 +/- 0.64	0.00153947 +/- 0.00004669	0.0003670
1.6e-250	Merina	Sumba	Mozambique_Chopi	33.81	23.32 +/- 0.69	0.00164251 +/- 0.00004452	0.0003852
3.8e-233	Merina	Dai	Esan	32.60	22.74 +/- 0.70	0.00176475 +/- 0.00005140	0.0004208

Table 6: MALDER results for all Malagasy populations, with varied African and Asian source populations. p value: significance of model tested. test pop: Malagasy target population. ref A: Asian population. ref B: African population. 2 ref z score: z score. 2 ref decay: Date, in generations before present, since admixture began with standard error. 2 ref amp exp: Amplitude of the LD decay curve. amp aff: Affine term estimating population substructure. *(continued)*

p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp	amp_aff
3.9e-244	Merina	Dai	Yoruba	33.37	22.75 +/- 0.68	0.00175932 +/- 0.00004984	0.0004199
1.8e-241	Merina	Dai	Mozambique_Chopi	33.19	23.29 +/- 0.70	0.00186752 +/- 0.00004541	0.0004395
5.5e-141	MerinaA	Borneo	Esan	25.28	21.70 +/- 0.86	0.00172501 +/- 0.00006674	0.0000510
2.5e-139	MerinaA	Borneo	Yoruba	25.13	21.69 +/- 0.86	0.00172051 +/- 0.00006620	0.0000510
1.8e-135	MerinaA	Borneo	Mozambique_Chopi	24.77	21.86 +/- 0.88	0.00180736 +/- 0.00006513	0.0000546
1.8e-128	MerinaA	Sumba	Esan	24.11	21.96 +/- 0.86	0.00151669 +/- 0.00006290	0.0000432
4.2e-129	MerinaA	Sumba	Yoruba	24.17	21.94 +/- 0.86	0.00151185 +/- 0.00006254	0.0000433
8.1e-137	MerinaA	Sumba	Mozambique_Chopi	24.90	22.17 +/- 0.89	0.00159416 +/- 0.00006327	0.0000466
1.4e-149	MerinaA	Dai	Esan	26.05	21.80 +/- 0.84	0.00173212 +/- 0.00006550	0.0000519
1.3e-149	MerinaA	Dai	Yoruba	26.05	21.80 +/- 0.84	0.00172795 +/- 0.00006479	0.0000519
1.2e-146	MerinaA	Dai	Mozambique_Chopi	25.79	21.98 +/- 0.85	0.00181659 +/- 0.00006355	0.0000556
2.3e-53	MerinaB	Borneo	Esan	15.38	26.42 +/- 1.72	0.00182359 +/- 0.00006434	0.0000235
1.2e-58	MerinaB	Borneo	Yoruba	16.15	26.46 +/- 1.64	0.00181557 +/- 0.00006173	0.0000232
4.2e-61	MerinaB	Borneo	Mozambique_Chopi	16.49	26.93 +/- 1.63	0.00193439 +/- 0.00005846	0.0000234
5.1e-52	MerinaB	Sumba	Esan	15.18	26.11 +/- 1.72	0.00159222 +/- 0.00006515	0.0000201
8.6e-57	MerinaB	Sumba	Yoruba	15.88	26.13 +/- 1.65	0.00158453 +/- 0.00006284	0.0000198
8.8e-60	MerinaB	Sumba	Mozambique_Chopi	16.31	26.75 +/- 1.64	0.00170122 +/- 0.00005956	0.0000200
1.9e-49	MerinaB	Dai	Esan	14.78	26.44 +/- 1.79	0.00181330 +/- 0.00006328	0.0000236
4.9e-54	MerinaB	Dai	Yoruba	15.48	26.48 +/- 1.71	0.00180611 +/- 0.00006077	0.0000233
3.8e-57	MerinaB	Dai	Mozambique_Chopi	15.93	26.94 +/- 1.69	0.00192369 +/- 0.00005553	0.0000236
6.2e-63	Mikea	Borneo	Esan	16.74	24.43 +/- 1.46	0.00179737 +/- 0.00007812	0.0000009
4.2e-61	Mikea	Borneo	Yoruba	16.49	24.45 +/- 1.48	0.00179523 +/- 0.00007745	0.0000008
8.6e-54	Mikea	Borneo	Mozambique_Chopi	15.44	23.99 +/- 1.55	0.00184531 +/- 0.00007972	0.0000014
6.7e-59	Mikea	Sumba	Esan	16.18	24.75 +/- 1.53	0.00158879 +/- 0.00007863	0.0000010
1.3e-56	Mikea	Sumba	Yoruba	15.86	24.77 +/- 1.56	0.00158630 +/- 0.00007822	0.0000009
6.3e-50	Mikea	Sumba	Mozambique_Chopi	14.86	24.30 +/- 1.64	0.00163606 +/- 0.00008013	0.0000015
2.2e-58	Mikea	Dai	Esan	16.11	24.68 +/- 1.53	0.00181233 +/- 0.00008078	0.0000010
1.1e-56	Mikea	Dai	Yoruba	15.87	24.70 +/- 1.56	0.00181012 +/- 0.00008012	0.0000009
2.4e-50	Mikea	Dai	Mozambique_Chopi	14.92	24.24 +/- 1.62	0.00186109 +/- 0.00008254	0.0000015
1.1e-70	Sakalava_Tsimihety	Borneo	Esan	17.77	22.42 +/- 1.26	0.00177727 +/- 0.00007455	0.0000396
4.3e-69	Sakalava_Tsimihety	Borneo	Yoruba	17.57	22.33 +/- 1.27	0.00176631 +/- 0.00007391	0.0000395
3.3e-70	Sakalava_Tsimihety	Borneo	Mozambique_Chopi	17.71	23.09 +/- 1.30	0.00188537 +/- 0.00007484	0.0000422
1.4e-73	Sakalava_Tsimihety	Sumba	Esan	18.14	22.70 +/- 1.25	0.00157199 +/- 0.00007657	0.0000344

Table 6: MALDER results for all Malagasy populations, with varied African and Asian source populations. p value: significance of model tested. test pop: Malagasy target population. ref A: Asian population. ref B: African population. 2 ref z score: z score. 2 ref decay: Date, in generations before present, since admixture began with standard error. 2 ref amp exp: Amplitude of the LD decay curve. amp aff: Affine term estimating population substructure. *(continued)*

p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp	amp_aff
3.5e-71	Sakalava_Tsimihety	Sumba	Yoruba	17.84	22.60 +/- 1.27	0.00156087 +/- 0.00007598	0.0000343
1.2e-73	Sakalava_Tsimihety	Sumba	Mozambique_Chopi	18.15	23.48 +/- 1.29	0.00167853 +/- 0.00007755	0.0000367
1.4e-72	Sakalava_Tsimihety	Dai	Esan	18.02	22.60 +/- 1.25	0.00177662 +/- 0.00007776	0.0000398
3.9e-71	Sakalava_Tsimihety	Dai	Yoruba	17.83	22.51 +/- 1.26	0.00176524 +/- 0.00007692	0.0000396
5.3e-72	Sakalava_Tsimihety	Dai	Mozambique_Chopi	17.94	23.26 +/- 1.30	0.00188434 +/- 0.00007879	0.0000423
9.6e-133	Temoro	Borneo	Esan	24.52	29.11 +/- 0.86	0.00194337 +/- 0.00007926	0.0000182
1e-128	Temoro	Borneo	Yoruba	24.14	29.07 +/- 0.90	0.00193347 +/- 0.00008011	0.0000178
7.4e-147	Temoro	Borneo	Mozambique_Chopi	25.81	29.35 +/- 0.87	0.00205756 +/- 0.00007973	0.0000186
5.3e-104	Temoro	Sumba	Esan	21.66	29.34 +/- 0.87	0.00172607 +/- 0.00007970	0.0000165
2.7e-101	Temoro	Sumba	Yoruba	21.37	29.28 +/- 0.90	0.00171615 +/- 0.00008032	0.0000162
7.5e-112	Temoro	Sumba	Mozambique_Chopi	22.47	29.60 +/- 0.86	0.00183343 +/- 0.00008158	0.0000169
5.4e-130	Temoro	Dai	Esan	24.26	29.48 +/- 0.82	0.00195981 +/- 0.00008079	0.0000181
6.1e-127	Temoro	Dai	Yoruba	23.97	29.42 +/- 0.85	0.00194883 +/- 0.00008131	0.0000178
2.8e-142	Temoro	Dai	Mozambique_Chopi	25.40	29.71 +/- 0.81	0.00207391 +/- 0.00008166	0.0000185
1.9e-69	Vezo	Borneo	Esan	17.62	21.98 +/- 1.25	0.00184944 +/- 0.00006413	0.0000285
5.5e-72	Vezo	Borneo	Yoruba	17.94	22.08 +/- 1.23	0.00184557 +/- 0.00006245	0.0000284
1.2e-72	Vezo	Borneo	Mozambique_Chopi	18.03	22.26 +/- 1.24	0.00194431 +/- 0.00005711	0.0000301
2.4e-67	Vezo	Sumba	Esan	17.34	22.16 +/- 1.28	0.00163475 +/- 0.00005995	0.0000246
1.6e-70	Vezo	Sumba	Yoruba	17.76	22.26 +/- 1.25	0.00163029 +/- 0.00005825	0.0000245
1.3e-70	Vezo	Sumba	Mozambique_Chopi	17.76	22.41 +/- 1.26	0.00172420 +/- 0.00005415	0.0000261
4.2e-66	Vezo	Dai	Esan	17.17	22.04 +/- 1.28	0.00185290 +/- 0.00006451	0.0000280
1.1e-68	Vezo	Dai	Yoruba	17.51	22.13 +/- 1.26	0.00184873 +/- 0.00006279	0.0000278
3.5e-70	Vezo	Dai	Mozambique_Chopi	17.71	22.29 +/- 1.26	0.00194813 +/- 0.00005747	0.0000296

Table 7: Pairwise FST values between all Malagasy populations.

	Betsileo	Betsimisaraka	Diego.Mixed	Merina.A	Merina.B	Mikea	Sakalava.Tsimihety	Temoro	Vezo
Betsileo	NA	0.0069522	0.0114288	0.0110161	0.0082805	0.0117686	0.0072215	0.0070768	0.0075091
Betsimisaraka	0.0069522	NA	0.0027162	0.0254780	0.0053308	0.0063718	0.0015886	0.0016943	0.0033650
Diego Mixed	0.0114288	0.0027162	NA	0.0335991	0.0070089	0.0069401	0.0018644	0.0029681	0.0043794
Merina A	0.0110161	0.0254780	0.0335991	NA	0.0255723	0.0309598	0.0261779	0.0256050	0.0246308
Merina B	0.0082805	0.0053308	0.0070089	0.0255723	NA	0.0103931	0.0056181	0.0052702	0.0069354
Mikea	0.0117686	0.0063718	0.0069401	0.0309598	0.0103931	NA	0.0062564	0.0065102	0.0048565
Sakalava/Tsimihety	0.0072215	0.0015886	0.0018644	0.0261779	0.0056181	0.0062564	NA	0.0016993	0.0029689
Temoro	0.0070768	0.0016943	0.0029681	0.0256050	0.0052702	0.0065102	0.0016993	NA	0.0034909
Vezo	0.0075091	0.0033650	0.0043794	0.0246308	0.0069354	0.0048565	0.0029689	0.0034909	NA

Alexander, David H., John Novembre, and Kenneth Lange. 2009. “Fast Model-Based Estimation of Ancestry in Unrelated Individuals.” *Genome Research* 19 (9): 1655–64. https://doi.org/10.1101/gr.094052.109.

Carter, Evelyn J., Eirlys E. Tysall, Jason A. Hodgson, and Andrea Manica. 2025. “Tidypopgen: Tidy Population Genetics in R.” *Methods in Ecology and Evolution* 17 (2). https://doi.org/10.1111/2041-210x.70204.

Carter, Evelyn, and Andrea Manica. 2025. “Tidypopgen: Tidy Population Genetics. R Package Version 0.0.0.9022.” https://evolecolgroup.github.io/tidypopgen.

Frazer, Kelly A., Dennis G. Ballinger, David R. Cox, David A. Hinds, Laura L. Stuve, Richard A. Gibbs, John W. Belmont, et al. 2007. “A Second Generation Human Haplotype Map of over 3.1 Million SNPs.” *Nature* 449 (7164): 851–61. https://doi.org/10.1038/nature06258.

Hodgson, Jason. 2016. “A Genomic Investigation of the Malagasy Confirms the Highland–Coastal Divide, and the Lack of Middle Eastern Gene Flow.” In, 231–54. https://doi.org/10.1007/978-3-319-33822-4_10.

Kent, W. James, Charles W. Sugnet, Terrence S. Furey, Krishna M. Roskin, Tom H. Pringle, Alan M. Zahler, and and David Haussler. 2002. “The Human Genome Browser at UCSC.” *Genome Research* 12 (6): 996–1006. https://doi.org/10.1101/gr.229102.

Loh, Po-Ru, Mark Lipson, Nick Patterson, Priya Moorjani, Joseph K Pickrell, David Reich, and Bonnie Berger. 2013. “Inferring Admixture Histories of Human Populations Using Linkage Disequilibrium.” *Genetics* 193 (4): 1233–54. https://doi.org/10.1534/genetics.112.147330.

Maier, Robert, and Nick Patterson. 2024. “Admixtools: Inferring Demographic History from Genetic Data. R Package Version 2.0.4.” https://github.com/uqrmaie1/admixtools.

Manichaikul, Ani, Josyf C. Mychaleckyj, Stephen S. Rich, Kathy Daly, Michèle Sale, and Wei-Min Chen. 2010. “Robust Relationship Inference in Genome-Wide Association Studies.” *Bioinformatics* 26 (22): 2867–73. https://doi.org/10.1093/bioinformatics/btq559.

Pickrell, Joseph K., Nick Patterson, Po-Ru Loh, Mark Lipson, Bonnie Berger, Mark Stoneking, Brigitte Pakendorf, and David Reich. 2014. “Ancient West Eurasian Ancestry in Southern and Eastern Africa.” *Proceedings of the National Academy of Sciences* 111 (7): 2632–37. https://doi.org/10.1073/pnas.1313787111.

Privé, Florian, Hugues Aschard, Andrey Ziyatdinov, and Michael G B Blum. 2018. “Efficient Analysis of Large-Scale Genome-Wide Data with Two R Packages: Bigstatsr and Bigsnpr.” *Bioinformatics* 34 (16): 2781–87. https://doi.org/10.1093/bioinformatics/bty185.

Weir, B. S., and C. Clark Cockerham. 1984. “Estimating F-Statistics for the Analysis of Population Structure.” *Evolution* 38 (6): 1358–70. https://doi.org/10.2307/2408641.